

Incentive-Compatible Information Design*

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Abstract

We study the design of mechanisms by an intermediary that generates information for a sender and also controls its disclosure to persuade a receiver about an unknown attribute of the sender. The sender is initially privately, but imperfectly, informed about her attribute, and the receiver takes an action based on posterior beliefs about the sender’s attribute and the sender’s belief about the attribute. We characterize incentive-compatible mechanisms when the sender contracts with the intermediary and specialize the analysis to study profit-maximizing mechanisms in two applications: college-admissions tests and digital market platforms.

1 Introduction

In recent decades, economic theory has developed sophisticated tools to study the extent to which information provision can affect the behavior and interaction of economic agents. Yet much of the attention has been focused on scenarios where the provision of information is through commitment, an omniscient (information) designer, or other exogenous considerations. This precludes environments where agents have to acquire information, often at a cost and from an intermediary. We study a general class of problems in which an intermediary sells an *information policy* to a privately but imperfectly informed sender who will interact *ex post* with a receiver. The allocation mechanism must be incentive-compatible, which restricts what information is available to the sender. The design of information subject to incentive compatibility of the allocation mechanism is the pervasive theme of this

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article. We specialize our theoretical framework to study profit-maximizing information policies in two applications: college-admissions tests and digital market platforms.

In our first application, we consider a firm that designs a college admissions test and sets a price to students for taking the test. The design of the test can reveal more or less information about the student's ability. Variations in the informativeness of the test affect how the college uses the test in admissions decisions, and this affects the student's willingness to pay for the test. In addition to these tradeoffs, the students have prior private (but imperfect) information about their ability and this translates into private information about how they will perform on the test and thus how much they are willing to pay for the test. The design of the test must therefore grapple with screening frictions.

Our main application concerns the design of a data policy by a digital platform. The platform can use the data it collects about users to estimate consumers' values for new products. This information is useful both to the consumers on the platform (to improve their purchase decisions) and to sellers on the platform (to target buyers and set prices). The platform can, for example, offer data management policies to consumers which help consumers make informed purchases and which also incentivize firms to set prices that maintain high consumer surplus. Consumers are willing to pay for such policies, and this can be a source of revenue for the platform. At the same time the consumers have their own private information about willingness to pay. The value of any data-management policy will be different for consumers with different private beliefs and so the platform faces screening frictions similar to those of the test-design firm.

These two applications belong to a broader class of environments in which a privately informed sender acquires information from a profit-maximizing intermediary to influence a receiver's decisions. The intermediary can design a policy that enhances the information of the sender while at the same time selectively disclosing information to the receiver to influence her ex-post action. We combine tools of mechanism design and information design to characterize incentive-compatible information policies coupled with obedient decision rules for the receiver.

The design of an optimal mechanism is inherently a multi-dimensional screening problem (for example, how finely to discriminate between students in the upper tail versus students in the lower tail). We provide the necessary conditions for a mechanism to be incentive compatible in terms of monotonicity of a functional of the information policy (equivalent to convexity of the sender's indirect utility) prescribed by the mechanism. These conditions are also sufficient when the sender's payoff is linear in her type.

In our college admission application, the student's utility is independent of her ability, as she only values being admitted to the college regardless of her ability.

The school (receiver) chooses to admit or reject the student based on its expectation of the student's ability. We use our characterization of incentive compatibility to enumerate the extreme points of the set of incentive-compatible, obedient, and individually rational information policies. We then use these extreme points to give a qualitative description of the optimal mechanism. We characterize the structure of the optimal mechanism under two conditions on the distribution F of students' private beliefs. Under the first condition, the optimal mechanism provides more informative (harder) tests to higher ability students and leaves rents only to the higher types. Whereas under the second condition, the optimal mechanism spreads out the informativeness (difficulty) of the tests more evenly among different types of students and leaves rents to all unexcluded types. Additionally, we give a sufficient condition on the distribution of senders' private beliefs under which the monotonicity constraints from incentive compatibility can be relaxed when both the sender and receiver have utilities linear in their posterior beliefs.

On the platform, a buyer may decline to purchase after receiving information about her value. We study mechanisms in which purchases at the recommended price are enforceable, but every truthful buyer is willing to purchase after receiving her information. Under this convention, incentive compatibility again has an envelope representation and a monotonicity condition.

The seller's obedience presents a further difficulty. Unlike the college, the seller responds to the entire distribution of posterior beliefs through the demand curve. We show that, conditional on each equilibrium price recommendation, additional private information can transform this distribution into a member of the unit-elastic family without changing platform participation, fees, or revenue. This gives a convex-order characterization of the allocations that can be supported by an obedient seller.

We first relax the convex-order constraint to a constraint on the conditional expectation and on the mass of marginal participating buyer types. Under a non-increasing weighted inverse-hazard condition, we obtain the optimal allocation of this relaxed problem. We then give additional sufficient conditions (the density is positive and continuously differentiable, and sufficiently close to a uniform distribution) under which the optimal relaxed mechanism solves the original single-price problem. At the optimal platform price, additional private information makes the seller willing to follow the recommendation while preserving allocations, fees, and revenue.

2 Model

Setup A sender has attribute $\theta \in \Theta = \{0, 1\}$, unknown to all. The sender is privately but imperfectly informed of θ as represented by her prior belief μ . Only

the sender knows her prior μ , and we will refer to μ as the type of the sender.

There is a receiver who is initially uninformed about the sender's attribute θ and her type μ , knowing only that the latter is distributed according to a prior distribution function F which is continuous and has density f .

An intermediary will design an information policy which will generate new information about θ . The intermediary can reveal some of this information to the sender, allowing the sender to update her belief to a posterior ν . The intermediary can also reveal information to the receiver. This will allow the receiver to form a *second-order* belief, i.e. an element of $\Delta(\Theta \times \Delta\Theta)$, about both the attribute θ and the posterior belief ν of the sender.

After observing the information provided by the intermediary, the receiver will take an action $a \in A$ and obtain a payoff $\pi_R(a, \beta)$. The sender will earn payoff $\pi_S(a, \nu)$.

Examples In the college admissions illustration the sender is a student and the receiver is a college. The intermediary sells a test to the student which will reveal information to the college to be used in admissions decision. Here θ is the student's ability, high ($\theta = 1$) or low ($\theta = 0$). The college chooses from the pair of actions $A = \{\text{admit}, \text{reject}\}$ and wants to admit students of high ability. Specifically the *ex-post* payoff from admitting the student depends only on θ and is given by the following matrix where $p \in (0, 1)$.

	$\theta = 0$	$\theta = 1$
admit	-1	$\frac{1-p}{p}$
reject	0	0

Given a belief β the college makes its admissions decision to maximize its expected payoff. In particular the student is admitted if and only if the college believes with at least probability p that the student has high ability. The student earns a payoff of 1 from being admitted and 0 from being rejected, independent of θ .

To avoid trivialities we assume $E_F \mu < p$ so that some information is necessary to induce the college to admit the student.

Our main application is to digital platforms. Here the sender is a buyer whose willingness to pay for a product is θ . The receiver is the seller of the product; he offers a price p which the buyer will either accept or reject. Rejection leads to a payoff of zero for both parties and acceptance leads to payoff $\theta - p$ for the buyer and p for the seller. Thus, when the buyer has posterior ν , she will accept any price $p \leq \nu$ and her total payoff will be $\max\{\nu - p, 0\}$. Given belief β , the seller's payoff from offering price p is given by

$$p \cdot \text{Prob}(\nu \geq p).$$

Note that the seller's payoff depends on his second-order belief: his belief about the buyer's belief.

Information Policies A *test* is defined by two non-empty sets of messages ("test results") $\mathcal{M}_S$ and $\mathcal{M}_R$ and for each θ a distribution $\rho_\theta \in \Delta(\mathcal{M}_S \times \mathcal{M}_R)$. When the sender submits to a test, the results are drawn from ρ_θ conditional on the sender's true attribute. The result in $\mathcal{M}_S$ is privately disclosed to the sender and the result in $\mathcal{M}_R$ is privately disclosed to the receiver. An information policy, or a *mechanism* is defined by a non-empty set of messages $\mathcal{M}_\Sigma$ and a rule that specifies as a function of $\mathcal{M}_\Sigma$ both a payment made by the sender and a test. The sender selects a message from $\mathcal{M}_\Sigma$ and the resulting test is carried out. Importantly, we restrict payments to be *ex ante*, i.e. before the realization of the test.

In a *direct* mechanism $\mathcal{M}_\Sigma = \Delta\Theta$, $\mathcal{M}_R = A$ and $\mathcal{M}_S = \Delta\Theta$. The sender reports her type $\mu \in \Delta\Theta$ and in response the intermediary charges the sender the fee $\phi(\mu)$, uses $\rho_\theta(\mu)$ to recommend an action $a \in A$ to the receiver and privately informs the sender of her posterior $\nu \in \Delta\Theta$. Thus, $\rho_\theta(\mu) \in \Delta(A \times \Delta\Theta)$. When the intermediary uses a direct mechanism and the realized recommendation is a , the receiver formulates the conditional belief $\beta(\cdot | a) \in \Delta(\Theta \times \Delta\Theta)$.

A direct mechanism is *obedient* if the recommended action is always the one that maximizes the receiver's payoff conditional on hearing the recommendation, i.e. with probability 1

$$a \in \operatorname{argmax}_{a' \in A} \pi_R(a', \beta(\cdot | a)).$$

Let $\tilde{v}(\mu) = \mu \operatorname{marg}_{\Delta\Theta} \rho_1(\mu) + (1 - \mu) \operatorname{marg}_{\Delta\Theta} \rho_0(\mu)$ be the total probability distribution of the posterior induced for type μ . A direct mechanism is *bayes-plausible* if for every μ ,

$$\mathbf{E} \tilde{v}(\mu) = \mu.$$

Recall that [Kamenica and Gentzkow \(2011\)](#) bayes-plausibility implies that for a sender with prior μ , the conditional probability of $\theta = 1$ conditional on every suggested posterior ν is ν itself. On the other hand, should μ report some other type $\mu' \neq \mu$ to the mechanism and receive the posterior ν suggested for μ' , she would use that suggestion to update her true prior μ and thereby obtain a possibly different posterior, call it $\delta_{\mu, \mu'}(\nu)$. By continuity of Bayesian updating, $\delta_{\mu, \mu'}(\nu)$ is continuous in μ for any given μ' and ν .

A direct mechanism yields *gross utility* for type μ equal to $v(\mu) = \mathbf{E}_{\rho(\mu)} \pi_S(a, \nu)$. We define the *indirect utility* function of a direct mechanism by

$$U(\mu) = v(\mu) - \phi(\mu)$$

When μ misreports $\mu' \neq \mu$ she instead earns gross utility

$$v(\mu' | \mu) = \mu \mathbf{E}_{\rho_1(\mu')} \pi_S(a, \delta_{\mu, \mu'}(\nu)) + (1 - \mu) \mathbf{E}_{\rho_0(\mu')} \pi_S(a, \delta_{\mu, \mu'}(\nu))$$

A direct mechanism is *incentive compatible* if for every μ, μ'

$$U(\mu) \geq v(\mu' | \mu) - \phi(\mu').$$

A direct mechanism is *individually rational* if $U(\mu) \geq U^*$ for all μ , where U^* is the sender's outside option from non-participating in the mechanism.

The goal of the intermediary is to maximize expected revenue. By the revelation principle we can restrict attention to obedient, bayes-plausible and incentive-compatible direct mechanisms (ρ, ϕ) designed to maximize

$$\Pi = \int_0^1 \phi(\mu) dF(\mu)$$

3 Characterization of Incentive Compatibility

The gross deviation utility $v(\mu' | \mu)$ is non-linear which complicates the usual envelope representation of incentive compatible mechanisms. However we can make use of the following observation to obtain a condition that is necessary in general and sufficient in some special cases.

Let (ρ, ϕ) be an incentive-compatible mechanism. Suppose for every μ' there exist numbers $q_0(\mu')$ and $q_1(\mu')$ such that for every μ

$$v(\mu' | \mu) \geq \mu q_1(\mu') + (1 - \mu) q_0(\mu') \quad (1)$$

with equality when $\mu = \mu'$. Then the linear function on the right-hand side is a support function for the indirect utility function:

$$\begin{aligned} U(\mu) &\geq v(\mu' | \mu) - \phi(\mu') \\ &\geq \mu q_1(\mu') + (1 - \mu) q_0(\mu') - \phi(\mu') \\ &= (\mu - \mu') q_1(\mu') + [(1 - \mu) - (1 - \mu')] q_0(\mu') \\ &\quad + \mu' q_1(\mu') + (1 - \mu') q_0(\mu') - \phi(\mu') \\ &= U(\mu') + (\mu - \mu') [q_1(\mu') - q_0(\mu')] \end{aligned}$$

This implies that $U(\cdot)$ is convex and hence absolutely continuous. It is differentiable at almost every μ with slope $q_1(\mu) - q_0(\mu)$. The indirect utility of type μ can be expressed as

$$U(\mu) = U(0) + \int_0^\mu [q_1(\mu') - q_0(\mu')] d\mu'$$

Turning things around, suppose that a mechanism (ρ, ϕ) yields a convex indirect utility function U and the inequality in [Equation 1](#) holds with equality for all μ . If the integral representation above holds then the mechanism is incentive compatible because by convexity.

$$U(\mu) \geq U(\mu') + (\mu - \mu') [q_1(\mu') - q_0(\mu')] = v(\mu' | \mu) - \phi(\mu').$$

4 Linear Problems

A first step to solving for an optimal mechanism in a general setting is solving the design problem in linear environments. We rely on two forms of linearity. We say the environment is *sender-linear* if

$$\pi_S(a, v) = v\pi_S(a, 1) + (1 - v)\pi_S(a, 0)$$

and an environment is *receiver-linear* if the receiver's payoff from any given action only depends on his first order beliefs about θ

$$\pi_R(a, \beta(\cdot | a)) = \pi_R(a, \text{marg}_{\Theta} \beta(\cdot | a))$$

We call the above preference linear as the receiver's payoff depends on $\beta(\cdot | a)$ only through $\text{marg}_{\Theta} \beta(\cdot | a) = \mathbf{E} \text{marg}_{\Delta\Theta} \beta(v | a)$

4.1 Example: College Admissions

College admissions illustrates the linear analysis: the sender has state-independent preferences and the receiver makes a binary choice based only on expected ability.

In the college admissions application a direct mechanism recommends admit or reject to the college and the obedience constraint is

$$\text{Prob}(\theta = 1 | \alpha = \text{admit}) \geq p$$

Let $q_{\theta}(\mu) = \text{marg}_A \rho_{\theta}(\mu) [\text{admit}]$ be the probability that a student with type μ and ability θ is recommended for admission. This environment is *sender-linear* and the gross deviation utility is

$$v(\mu' | \mu) = \mu q_1(\mu') + (1 - \mu)q_0(\mu')$$

for all μ and μ' , and therefore the monotonicity condition, namely for all $\mu \geq \mu'$

$$q_1(\mu) - q_0(\mu) \geq q_1(\mu') - q_0(\mu')$$

together with the envelope formula are necessary and sufficient conditions for incentive compatibility.

For an optimal incentive-compatible mechanism there must exist a type μ^0 whose participation constraint binds, i.e. $U(\mu^0) = 0$. If not, fees could be increased by a constant for all types, raising revenue without altering incentive constraints.

Then we may write

$$U(\mu) = \begin{cases} - \int_{\mu}^{\mu^0} [q_1(\mu') - q_0(\mu')] d\mu', & \text{if } \mu \leq \mu^0 \\ \int_{\mu^0}^{\mu} [q_1(\mu') - q_0(\mu')] d\mu', & \text{if } \mu \geq \mu^0 \end{cases}$$

Since U is convex and weakly positive, the slope must be non-positive below μ^0 and non-negative above. In particular $q_1(\mu) \leq q_0(\mu)$ for all $\mu \leq \mu^0$.

We may express the intermediary's expected revenue as expected gross utility minus expected indirect utility.

$$\begin{aligned} \Pi = \int_0^{\mu^0} \left\{ v(\mu) + \int_{\mu}^{\mu^0} [q_1(\mu') - q_0(\mu')] d\mu' \right\} f(\mu) d\mu \\ + \int_{\mu^0}^1 \left\{ v(\mu) - \int_{\mu^0}^{\mu} [q_1(\mu') - q_0(\mu')] d\mu' \right\} f(\mu) d\mu \end{aligned}$$

We can see immediately that profit is increasing in $q_1(\mu)$ for all $\mu \leq \mu^0$. Moreover, increasing $q_1(\mu)$ adds slack to the obedience constraint and therefore the constraint $q_1(\mu) \leq q_0(\mu)$ must bind and we have $q_1(\mu) = q_0(\mu)$ for all $\mu \leq \mu^0$.

It follows that $q_1(\mu) \geq q_0(\mu)$ for all $\mu \in [0, 1]$ and in fact we may take $\mu^0 = 0$, that is $U(0) = 0$.

When we change the order of integration and simplify we obtain the following expression for expected *virtual surplus*

$$\Pi = \mathbf{E} \left[q_1(\mu) \left(\mu - \frac{1 - F(\mu)}{f(\mu)} \right) \right] + \mathbf{E} \left[q_0(\mu) \left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right) \right] \quad (2)$$

The problem reduces to maximizing expected virtual surplus subject to monotonicity, obedience, and $q_1(\mu) \geq q_0(\mu)$.

Notice that the sum of the coefficients on $q_1(\mu)$ and $q_0(\mu)$ equals 1. If we ignore the obedience constraint, it would be optimal therefore to set $q_1(\mu) = q_0(\mu) = 1$ for all μ . This is the mechanism that admits the student with probability 1 (and extracts all of the surplus), but this mechanism violates obedience because $\mathbf{E}\mu < p$.

It follows that the obedience constraint is binding at the optimum and the problem becomes one of restoring obedience at minimum cost in terms of foregone virtual surplus, and subject to monotonicity and $q_1(\mu) \geq q_0(\mu)$. Plugging in the binding obedience constraint

$$\text{Prob}(\theta = 1 \mid \alpha = \text{admit}) = p$$

in [Equation 2](#) we obtain that

$$\Pi = \underbrace{\mathbf{E} \left[\frac{\mu}{p} q_1(\mu) \right]}_{\text{Information design tradeoff}} - \underbrace{\mathbf{E} \left[\frac{1 - F(\mu)}{f(\mu)} (q_1(\mu) - q_0(\mu)) \right]}_{\text{Distortion from screening}}$$

The first term represents the sender's tradeoff in a standard persuasion setting, where the sender can commit to a costless test. The second term represents how

the designer distorts the mechanism to capture more information rents from the sender.

One candidate mechanism is a threshold mechanism in which $q_1(\mu) = q_0(\mu)$ for all μ and these admission probabilities jump from 0 to 1 at the interior point $\tilde{\mu}$ defined by

$$E(\mu \mid \mu \geq \tilde{\mu}) = p$$

This mechanism is feasible because it is monotonic, and satisfies both obedience and $q_1(\mu) \geq q_0(\mu)$. Moreover its indirect utility has a constant slope of zero so the mechanism extracts all of the surplus it generates. Nevertheless this mechanism is typically not optimal. Instead a mechanism which selectively rejects some low-ability students (by setting $q_1(\mu) > q_0(\mu)$ for some types), can generate larger revenues despite yielding positive rents to the student.

Proposition 1. *The optimal revenue is achieved by allocation of the following form, for some values $0 \leq \mu_0 \leq \bar{\mu} \leq 1$.*

1. If $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-increasing then optimal allocation (q_1, q_0) is of the form

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1, 1) & \mu \in [\mu_0, \bar{\mu}) \\ (1, 0) & \mu \in [\bar{\mu}, 1] \end{cases}$$

$$\text{Where } \int_{\mu_0}^1 \mu dF(\mu) = \frac{p}{1-p} \int_{\mu_0}^{\bar{\mu}} (1 - \mu) dF(\mu).$$

2. If $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-decreasing then optimal allocation (q_1, q_0) is of the form

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1 - \alpha, 0) & \mu \in [\mu_0, \bar{\mu}) \\ (1, \alpha) & \mu \in [\bar{\mu}, 1] \end{cases}$$

$$\text{Where } (1 - \alpha) \int_{\mu_0}^{\bar{\mu}} \mu dF(\mu) + \int_{\bar{\mu}}^1 \mu dF(\mu) = \alpha \frac{p}{1-p} \int_{\bar{\mu}}^1 (1 - \mu) dF(\mu).$$

The optimal mechanism obtained in [item 1](#) allocates informative tests to the higher types, and lower types are offered uninformative tests. Among students who are not excluded, those with lower prior expected ability take easier tests (less informative) and pay higher fees per student. Students with higher prior expected ability take difficult tests (more informative) and earn greater information rent. In contrast, the optimal mechanism obtained in [item 2](#) allocates informative tests more evenly among different applicant types. In fact, there is a reversal in the allocation of perfectly revealing tests. For [item 2](#), types in the middle are admitted only if the student is high-ability, and higher types are admitted even if they are low-ability.

Whereas in [item 1](#), types in the middle are always admitted and higher types are admitted only if they are high-ability. For a welfare perspective, all types that are not excluded by the intermediary earn information rents in [item 2](#), whereas the intermediary extracts all surplus from types that are allocated uninformative tests in [item 1](#).

General Linear Environments: As demonstrated by our example above, the key analytical difficulty in computing an optimal mechanism is the presence of both aggregate/moment constraints from obedience and point-wise monotonicity constraints from incentive-compatibility. When the environment is *sender-linear*, we can express the designer's revenue in terms of the expected virtual surplus as in [Equation 2](#)

$$\begin{aligned} \Pi = \mathbf{E} \left[\pi_S(a, 1) \left(\mu - \frac{1 - F(\mu)}{f(\mu)} \right) \rho_1(\mu)(a) \right] \\ + \mathbf{E} \left[\pi_S(a, 0) \left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right) \rho_0(\mu)(a) \right] - U(0) \end{aligned} \quad (3)$$

Along with the envelope representation, the following monotonicity condition is necessary and sufficient for incentive compatibility

$$K(\mu) = \mathbf{E} [\pi_S(a, 1) \rho_1(\mu)(a) - \pi_S(a, 0) \rho_0(\mu)(a)] \quad \text{is weakly increasing in } \mu. \quad (4)$$

The next result gives two conditions on F under which we can simplify the designer's optimization problem by relaxing the monotonicity of the slope of the sender's indirect utility ([Equation 4](#)).¹

Theorem 1. Suppose $f > 0$ on $(0, 1)$, $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is strictly decreasing, and $\frac{F(\mu)}{\mu f(\mu)}$ is strictly increasing. If the environment is sender-linear and receiver-linear, A is a compact metric space, and $\pi_S(\cdot, \theta)$ is continuous for each θ , then any obedient, individually rational direct mechanism that maximizes [Equation 3](#) subject to the envelope formula has an incentive-compatible version differing only on a set of types of F -measure zero.

The proof is in [Appendix B](#).

5 Sender's Ex-post Participation

As argued before, the *sender-linear* environment admits a rather straightforward characterization of incentive compatibility in terms of the envelope representation

¹At a revenue maximum some type μ^0 satisfies $U(\mu^0) = U^*$; otherwise all fees could be increased by a common amount. In for the binding type μ^0 , rents below this type are weighted by F , and rents above it by $1 - F$. The condition involving F handles types to the left of a binding type, and the weighted inverse-hazard condition handles those to the right.

and monotonicity of the slope of the sender's indirect utility. That said, linearity often fails in natural economic models; a particular failure of linearity is due to the sender's decision to participate ex-post. For example, a buyer might learn from the platform that the item is worthless and then decide not to trade at the platform's recommended price. In this situation, the sender's payoff is not linear in her type, as the sender can choose her outside option over the recommended action after observing the outcome of her chosen test. Formally, the sender's payoff takes the following form

$$\max\{ \pi_S(a, \nu), U^* \}$$

Where U^* is the sender's outside option. We allow the intermediary to **enforce** the sender's participation when the receiver accepts the recommended action. We nevertheless require that, after truthful reporting and the prescribed information, the sender weakly prefers participation to her outside option. We call this requirement *full participation*.² The resulting participation requirement is

$$\pi_S(a, \nu) \geq U^* \quad \text{for all } a \text{ and } \nu \text{ in support of } \bar{\rho}(\mu) \quad (5)$$

Where $\bar{\rho}(\mu) = \mu\rho_1(\mu) + (1 - \mu)\rho_0(\mu)$. The above requires that whenever a type μ sender, assuming truth telling, is recommended to action a and has a posterior ν , then she is weakly better off from participating than exercising her outside option. We now apply this participation requirement to the digital platform.

5.1 Example: Digital Platform

We now use the ideas developed so far to study the optimal design of digital platforms. A platform has access to detailed data which enables it to precisely estimate a buyer's willingness to pay for products. The platform can provide this information to the buyer but moreover to the seller. The former guides the buyer's purchases while the latter disciplines the prices set by sellers. The platform sells access to the buyer and by controlling the information available to sellers the platform also controls the consumer surplus provided to buyers and thus increases the value of membership on the platform. We will describe how this application maps to the general setting analyzed above.

Formally, the buyer plays the role of the sender, with willingness to pay θ , and private information μ . The seller is the receiver, setting a price $p \in A = [0, 1]$. A sale at price p yields $\theta - p$ for the buyer and p for the seller. The payoff to both from no sale (including when the buyer is excluded from the platform) is zero.

²The commitment applies after every report. A veto right would also allow the sender to reject the outcome after a misreport, and therefore changes the incentive constraints; see [Compte and Jehiel \(2009\)](#).

A mechanism for the platform specifies whether the buyer has access to the platform and if so provides information to the buyer leading to the posterior ν and recommends the price p to the seller. Moreover, it must be obedient for the seller to offer the price p we describe the obedience constraints further in [Section 6](#)

Participation Let $q_\theta(\mu)$ denote the probability of platform access for type μ with willingness to pay θ , and let

$$\bar{q}(\mu) = \mu q_1(\mu) + (1 - \mu) q_0(\mu)$$

be the total probability that μ joins the platform. The buyer earns non-negative payoff only when joining the platform. The participation constraint, [Equation 5](#), therefore requires that the expected payoff for every type μ conditional on joining the platform and paying price p is non-negative:

$$\mu \cdot q_1(\mu) \geq p \bar{q}(\mu) \quad (6)$$

Incentive Compatibility The following observation enables us to use our general representation of incentive compatibility. Since the buyer's payoff is zero when excluded from the platform this payoff is equivalent to the payoff from purchasing the good at price equal to her willingness to pay θ . Thus we may without loss represent any mechanism as one in which sale occurs with probability 1 at a random price from the set $\{0, p, 1\}$.

In particular the expected price paid by type μ having willingness to pay θ is

$$\bar{p}_1(\mu) = p \cdot q_1(\mu) + 1 \cdot (1 - q_1(\mu)) \quad (7)$$

for $\theta = 1$ and

$$\bar{p}_0(\mu) = p \cdot q_0(\mu) + 0 \cdot (1 - q_0(\mu)) \quad (8)$$

for $\theta = 0$. Since the total (both on and off the platform) expected value of the buyer's posterior is equal to her type μ , this yields gross payoff

$$v(\mu) = \mathbf{E}(\nu - p) = \mu - (1 - \mu) \bar{p}_0(\mu) - \mu \bar{p}_1(\mu)$$

where $\bar{p}_\theta(\mu)$ is the expected value of the random price offered to type μ , conditional on having true willingness to pay θ . The deviation payoff is

$$v(\mu' | \mu) = \mu - \mu \bar{p}_1(\mu') - (1 - \mu) \bar{p}_0(\mu') = \mu(1 - \bar{p}_1(\mu')) - (1 - \mu) \bar{p}_0(\mu')$$

and therefore the monotonicity condition requires that $1 - \bar{p}_1(\mu) + \bar{p}_0(\mu)$ is weakly increasing. Or using the expressions in [Equation 7](#) and [Equation 8](#),

$$K(\mu) = (1 - p)q_1(\mu) + pq_0(\mu) \quad \text{is weakly increasing in } \mu. \quad (9)$$

This condition together with the envelope formula

$$U(\mu) = \int_0^\mu [(1-p)q_1(\mu') + pq_0(\mu')] d\mu' + U(0) \quad (10)$$

are necessary and sufficient for incentive compatibility under the enforcement convention above.³

6 Obedience on the Digital Platform

In the college admissions example, obedience depends only on the college's posterior mean. On the platform, the seller trades off a higher price against a lower probability of sale, so obedience depends on the entire distribution of buyers' posterior beliefs. We now characterize this constraint directly for the platform.

Consider an equilibrium price recommendation $p \in (0, 1)$ conditional on platform access. To simplify on notation we will use the following for the conditional distribution of buyer beliefs

$$G = \text{marg}_{\Delta\Theta} \beta(\cdot \mid p)$$

Full participation requires support in $[p, 1]$. The seller earns p when the recommendation is followed, and obedience requires

$$p \geq p' \text{Prob}_G(v \geq p') \quad \text{for every } p' \in [0, 1]. \quad (11)$$

The weak inequality in the purchase decision includes buyers who are indifferent at the offered price. With a continuously distributed price recommendation, conditional statements are understood almost surely under its distribution.

A useful family of obedient distributions is

$$H_x^p(s) = \begin{cases} 0 & s < p, \\ \max\{x, 1 - \frac{p}{s}\} & p \leq s < 1, \\ 1 & s = 1, \end{cases} \quad x \in [0, 1]. \quad (12)$$

At $x = 0$, this is the unit-elastic distribution of [Roesler and Szentes \(2017\)](#): the seller is indifferent between all prices in $[p, 1]$. For $x > 0$, there is an atom at p and a flat part of the CDF immediately above it. This is the full-trade, zero-cost case of the canonical demands in [Chopra and Ely \(2026b\)](#), where additional private information shapes demand while preserving the equilibrium price and buyer and seller surplus. Every distribution in the family satisfies [Equation 11](#), including the

³The buyer cannot renege on the purchase when the seller accepts the recommended price p . If the seller chooses a different price, the buyer is free to decline. The resulting demand determines the seller's obedience constraint in [Section 6](#).

atom at one. Its mean decreases continuously from $\mathbf{E}H_0^p = p(1 - \ln p)$ to p as x increases from zero to one.

The family also provides a useful representation of any obedient platform mechanism.

Theorem 2. *Every obedient platform mechanism satisfying incentive compatibility, individual rationality, and full participation can be transformed, by providing additional private information to buyers, into a mechanism with the same state-contingent allocation of platform access and price recommendations, fees, and revenue, for which the conditional distribution of buyer beliefs is $H_{x(p)}^p$ at almost every recommended price $p \in (0, 1)$. Here $x(p) \in [0, 1]$ may depend on the recommendation.*

Price recommendations $p = 0$ and $p = 1$ are degenerate. At zero, seller obedience requires buyer belief zero; at one, full participation requires belief one. In both cases the buyer's gross surplus is zero. For the buyer's payoffs, these recommendations can therefore be interpreted as exclusion from the platform, as in the representation in [Equation 7–Equation 8](#).

[Theorem 2](#) gives a revelation-type representation of equilibrium outcomes: additional private information places the distribution of buyer beliefs in the iso-elastic family while preserving the allocation and payoffs. In what follows, we focus on implementing a platform with a single nontrivial price $p \in (0, 1)$.

To apply the result when choosing an allocation at a fixed price p , let G_q denote the conditional distribution of buyer beliefs induced by platform access when the allocation is q . The allocation, fee, and price determine payoffs, incentive compatibility, interim participation, and revenue under the enforcement convention. The platform can then provide additional information privately to buyers to shape their beliefs and discipline the seller's pricing decision, while ensuring that truthful buyers remain willing to purchase. At a type μ with platform access, the posterior before this additional information is

$$v_q(\mu) = \frac{\mu q_1(\mu)}{\bar{q}(\mu)}.$$

The law G_q weights types by $\bar{q}(\mu) dF(\mu)$, normalized by the probability of platform access. It summarizes the beliefs induced by the allocation; individual report incentives depend on the underlying allocation and fees.

Corollary 1. *A single-price allocation with positive probability of platform access and fees satisfying report incentive compatibility and participation can be implemented by an obedient information policy if and only if, for some $x \in [0, 1]$,*

$$H_x^p \succeq_{mps} G_q. \tag{13}$$

The information policy preserves the allocation, fees, and revenue.

Here $H \succeq_{\text{mps}} G$ means that G is a mean-preserving contraction of H : $\int \eta dH \geq \int \eta dG$ for every continuous convex function η . Both results are proved in [Appendix C](#). With universal exclusion there is no conditional belief distribution to normalize and obedience is vacuous.

The platform screens buyers through access and fees, while additional private information changes the demand curve faced by the seller, hence discipline her obedience. The convex-order condition identifies when an allocation can be supported by an obedient price without changing membership or its value.

7 Relaxing Platform Obedience

The convex-order condition in [Corollary 1](#) is a collection of integral inequalities. We first retain two necessary conditions and study the optimal allocation at a given price. The platform then chooses its price to obtain an optimal relaxed mechanism. We subsequently give a sufficient condition under which this mechanism can be implemented by an obedient information policy.

The first necessary condition is equality of conditional means:

$$\mathbf{E}G_q = \mathbf{E}H_x^p. \quad (14)$$

The second bounds the mass at the lowest posterior among participating buyers:

$$G_q(p) \leq x. \quad (15)$$

Indeed, if the latter inequality failed, the integrated CDF inequality would fail immediately above p . These conditions alone do not guarantee obedience; the additional information must satisfy [Equation 13](#).

Relaxed Problem When we express the platform's profit as the difference between expected gross payoff and expected buyer utility, represent expected utility by the envelope formula with $U(0) = 0$ and then proceed with the standard manipulations we obtain the following objective function for the platform.

$$\begin{aligned} \Pi = \int_0^1 (1-p) \left(\mu - \frac{1-F(\mu)}{f(\mu)} \right) q_1(\mu) dF(\mu) \\ - \int_0^1 p \left(1 - \mu + \frac{1-F(\mu)}{f(\mu)} \right) q_0(\mu) dF(\mu) \end{aligned} \quad (16)$$

Our *relaxed problem* is to choose $q_\theta(\cdot)$ to maximize profit subject to monotonicity ([Equation 9](#)), participation ([Equation 6](#)), and relaxed obedience ([Equation 14](#) and [Equation 15](#)). We call an allocation and its envelope payments satisfying these constraints a *relaxed mechanism*.

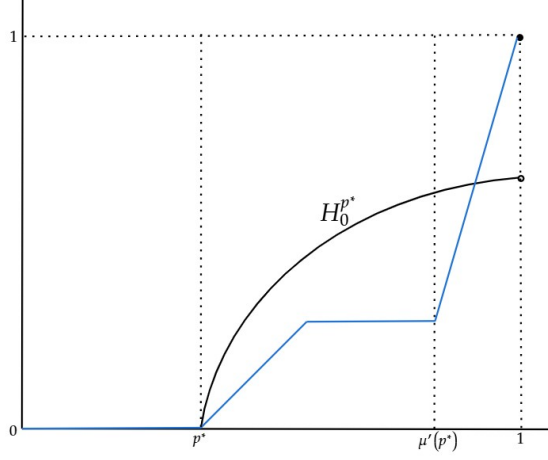


Figure 1: Schematic uniform illustration of the belief distribution induced by the allocation (blue) and the unit-elastic benchmark H_0^p . Additional private information implements the benchmark while preserving platform access and fees; the proof verifies the required integrated CDF inequalities.

Proposition 2. Suppose F has a positive C^1 density sufficiently close to uniform in the supremum norm and $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-increasing. There is an optimal relaxed mechanism whose allocation belongs to one of the following two families $q(\mu) = (q_1(\mu), q_0(\mu))$:

$$q(\mu) = \begin{cases} (0, 0) & \text{if } \mu < \mu_0 \\ \left(\frac{p}{1-p} \frac{1-\mu_0}{\mu_0}, 1\right) & \text{if } \mu \in [\mu_0, \mu'] \\ (1, 1) & \text{otherwise} \end{cases} \quad \text{or} \quad q(\mu) = \begin{cases} (0, 0) & \text{if } \mu < \mu_0 \\ (1, 0) & \text{if } \mu \in [\mu_0, \mu'] \\ (1, 1) & \text{otherwise} \end{cases}$$

for feasible thresholds $0 \leq \mu_0 \leq \mu' \leq 1$. We refer to the two displayed allocations as the first and second forms, respectively. In the first form, $\mu_0 \geq p$; participation in the second requires $\mu' \geq p$.

For each fixed price, the platform chooses allocations and fees, then optimizes over prices. The proof in [Section D.2–Section D.4](#) first holds the slope fixed to determine the allocation, then uses perturbations that preserve expected gross surplus and reduce information rents.

Among participating types, the likelihood ratio $q_1(\mu)/q_0(\mu)$ is one above μ' and constant on the middle interval. The distribution of beliefs induced by the allocation has no atom at p . The maximal-mean argument in [Section D.5](#) shows that a positive-revenue optimal relaxed mechanism of either displayed form has conditional mean $\mathbf{E}H_0^p$.

The next theorem establishes that the relaxed revenue bound is attainable in the original single-price problem. In addition to the weighted inverse-hazard restric-

tion, it requires the density of buyers' private beliefs to be positive, continuously differentiable, and uniformly close to one: equal-length intervals of types must contain nearly equal probabilities. Under these conditions, the beliefs induced by the optimal allocation can be spread into the unit-elastic distribution through additional private information. This makes the seller willing to follow the recommended price while preserving each type's platform access, fee, and expected payoff. The proof is in [Section D.6](#), after the discussion of double deviations.

Theorem 3. *There exists $\varepsilon > 0$ such that, if F has a positive C^1 density f on $[0, 1]$ satisfying $\|f - 1\|_\infty < \varepsilon$ and $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-increasing, an optimal relaxed mechanism can be implemented by an obedient information policy with the same allocation, fees, and revenue. Hence the relaxed problem and the original single-price problem have the same optimum. In particular, this holds for uniform.*

As the proof of [Theorem 3](#) suggests, the key difficulty is showing that an allocation of the first form in [Proposition 2](#) is a mean-preserving contraction of the iso-elastic distribution H_0^p as in general the resulting posterior distribution G_q can cross H_0^p at multiple points. The theorem first identifies a necessary condition on the range platform's optimal price for which an allocation of the second form does not dominate the revenue from the optimal first-form allocation. Then for this range of prices we verify the required convex order condition from [Corollary 1](#).

8 Conclusion and Open Problems

We develop tools for mechanisms in which a privately informed agent buys information from an intermediary to persuade a receiver. The intermediary jointly designs information and fees, subject to the agent's reporting incentives. These tools can also inform the design information policies in other application, for example a natural extension is to credit rating agencies that sell assessments to borrowers seeking better financing terms, incorporating the selection effects studied by [Chopra and Ely \(2026a\)](#).

Our analysis also raises two open problems. First, the platform solution relaxes obedience and then supplies additional private information to restore it. This preserves reporting incentives when purchases at the recommended price are enforceable. Without that commitment, a buyer may combine a misreport with a subsequent refusal to purchase; additional information can change the profitability of this double deviation. An optimal mechanism must guard against such *double deviations* of the buyer. A general treatment of participation and information provision across applications is open.

Second, our platform solution restricts recommendations to a single nontrivial price. It follows from the buyer-optimal benchmarks of [Roesler and Szentes \(2017\)](#)

and Chopra and Ely (2026b) that single-price implementations is without loss when reporting constraints are absent, and the platform can use type-dependent fees to extract the resulting buyer surplus. For uniformly distributed types and enforceable purchases, the two-price mechanism in the supplement (Chopra and Ely, 2026c, ??) is obedient, incentive-compatible, individually rational, and satisfies full participation. It earns approximately 0.06680, exceeding the revenue approximately 0.06522 obtained within the two single-price forms in the supplement (Chopra and Ely, 2026c, ??). Characterizing the optimal segmentation, and the conditions under which one nontrivial price suffices, remains open.

9 Related Literature

Corrao (2023) and Lizzeri (1999) study variations of an information design problem with incentive constraints but in their model the sender is already perfectly informed. The role of the intermediary is to turn the sender’s soft information into information that is verifiable to the receiver. In Ali et al. (2020) the intermediary is generating new information but the sender has no private information *ex ante* and so there are no screening frictions. Bergemann et al. (2018) study a problem like ours where the information designer produces new information for an *ex ante* privately informed agent. However, in Bergemann et al. (2018) the sender herself takes the action *ex post*, there is no third party/receiver.

Three concurrent and independent papers are closest to ours. Celik and Strausz (2025) and Weksler and Zik (2025) both study a buyer-seller interaction like our digital platform and Mäkimattila et al. (2024) study a version of our college admissions example. All three papers impose a restriction on the intermediary that the test chosen by the sender must be revealed to the receiver. By contrast we give the intermediary full flexibility in designing the information structure for the sender and the receiver. This ability to pool is valuable to the intermediary and shapes the optimal mechanism. As a benchmark Mäkimattila et al. (2024) do also examine the fully flexible case and derive the optimal mechanism in a case similar to our [item 1](#). Madarász and Pycia (2025) develop a cost-over-content result and apply it to a platform setting like ours. Their characterization differs as they allow the platform to charge transfers to the seller. Additionally, the authors also investigate variations of the college admission example in Chopra (2025) and Ely (2025). Ely (2025) studies optimal test allocation when the student can credibly reveal her private type to the college, leading to non-linear sender payoffs. Chopra (2025) studies the designer’s problem in a general linear environment with attribute-independent preferences for the sender, but importantly considers multiple competing receivers and the feedback between the receivers’ commitment and the optimal mechanism.

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A College Admissions

A.1 Geometry of Monotone Tests

The slope of the indirect utility is given by $K(\mu) = q_1(\mu) - q_0(\mu)$. Substituting this into [Equation 2](#) gives us

$$\Pi = \mathbf{E}[q_1(\mu)] - \mathbf{E}\left[K(\mu) \left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)}\right)\right]$$

The obedience constraint is given by

$$\mathbf{E}[\mu q_1(\mu)] \geq \frac{p}{1-p} \mathbf{E}[(1-\mu)q_0(\mu)]$$

As we argued in the text the obedience constraint must bind at an optimum. We can express the binding constraint in terms of the slope function $K(\mu)$ as follows.

$$\mathbf{E}[(1-\mu)K(\mu)] = \mathbf{E}\left[\left(1 - \frac{\mu}{p}\right) q_1(\mu)\right]$$

Putting together,

$$\Pi = \mathbf{E}\left[\frac{\mu}{p} q_1(\mu)\right] - \mathbf{E}\left[K(\mu) \frac{1 - F(\mu)}{f(\mu)}\right] \quad (17)$$

If K is such that there exists some incentive compatible and obedient test (q_1, q_0) for which $K(\mu) = q_1(\mu) - q_0(\mu)$, the following holds

$$0 \leq K(\mu) \leq q_1(\mu) \leq 1 \quad (18)$$

and

$$\mathbf{E}[(1-\mu)K(\mu)] \geq \int_0^1 \left(1 - \frac{\mu}{p}\right) q_1(\mu) dF(\mu) \quad (19)$$

We call a slope K feasible if it satisfies [Equation 18](#) and [Equation 19](#) for some test allocation q_1 . Define the following class of threshold allocations

$$(q_1^K(\mu), q_0^K(\mu)) := \begin{cases} (K(\mu), 0) & \mu < \mu^K \\ (1, 1 - K(\mu)) & \mu \geq \mu^K \end{cases}$$

For some $\mu^K \in [0, 1]$.

Lemma 1. For given p and feasible $K(\mu)$, the optimal allocation is $q_1 = q_1^K$ for some μ^K .

Proof. It is clear from [Equation 17](#) and [Equation 19](#) that the optimum requires $q_1(\mu) = 1$ for all $\mu \geq p$.

Consider for contradiction that there is some interval of types $[\mu_0, \bar{\mu})$ such that $K(\mu) < q_1(\mu) < 1$. It's sufficient to consider $\bar{\mu} \leq p$. Pick $\delta > 0$ such that $\bar{\mu} - \mu_0 \geq 2\delta$. We can improve the revenue by increasing q_1 on $[\bar{\mu} - \delta, \bar{\mu})$ by some $\varepsilon_1 > 0$ and by reducing q_1 on $[\mu_0, \mu_0 + \delta)$ by $\varepsilon_0 > 0$ where

$$\varepsilon_1 = \varepsilon_0 \frac{\int_{\mu_0}^{\mu_0 + \delta} \left(1 - \frac{\mu}{p}\right) dF(\mu)}{\int_{\bar{\mu} - \delta}^{\bar{\mu}} \left(1 - \frac{\mu}{p}\right) dF(\mu)}$$

ensuring that [Equation 19](#) is maintained.

The revenue from the new allocation is greater than the old allocation if the following holds:

$$\frac{\int_{\mu_0}^{\mu_0 + \delta} \left(1 - \frac{\mu}{p}\right) dF(\mu)}{\int_{\bar{\mu} - \delta}^{\bar{\mu}} \left(1 - \frac{\mu}{p}\right) dF(\mu)} \geq \frac{\int_{\mu_0}^{\mu_0 + \delta} \frac{\mu}{p} dF(\mu)}{\int_{\bar{\mu} - \delta}^{\bar{\mu}} \frac{\mu}{p} dF(\mu)} \iff \frac{\int_{\mu_0}^{\mu_0 + \delta} dF(\mu)}{\int_{\bar{\mu} - \delta}^{\bar{\mu}} dF(\mu)} \geq \frac{\int_{\mu_0}^{\mu_0 + \delta} \mu dF(\mu)}{\int_{\bar{\mu} - \delta}^{\bar{\mu}} \mu dF(\mu)}$$

The last inequality follows from our choice of δ .

Now consider q_1 such that there are intervals $I_1 < I_2$ where $q_1(\mu) = K(\mu)$ for $\mu \in I_2$ and $q_1(\mu) = 1$ for $\mu \in I_1$. It suffices to consider $\sup(I_2) \leq p$ by the previous arguments. We construct an improvement similar to the above by slightly increasing q_1 on I_2 and reducing q_1 on I_1 to preserve the inequality in [Equation 19](#). The revenue of this improvement is greater than the old allocation if the following holds:

$$\frac{\int_{I_1} dF(\mu)}{\int_{I_2} dF(\mu)} \geq \frac{\int_{I_1} \mu dF(\mu)}{\int_{I_2} \mu dF(\mu)}$$

This is implied by $\frac{\inf(I_2)}{\sup(I_1)} \geq 1$. □

By [Lemma 1](#) we can, without loss, focus on allocations that have non-decreasing admission probability q_1 . Let Λ be the set of all such allocations which are also incentive-compatible and have indirect utility with $U(0) = 0$. When viewed as a subset of the topological vector space (TVS) $\mathbb{L}_\infty([0, 1] \rightarrow \mathbb{R}^2)$, the set Λ is algebraically compact and convex.⁴ Moreover, its extreme points are such that for some $0 \leq \mu_0 \leq \bar{\mu} \leq 1$

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1, 1) & \mu \in [\mu_0, \bar{\mu}) \\ (1, 0) & \mu \in [\bar{\mu}, 1] \end{cases}$$

⁴A set Λ in a TVS is algebraically compact if the intersection of the set Λ with a line is always algebraically closed and bounded, see [Barvinok \(2025\)](#).

The set of such allocations is $\text{ex}(\Lambda)$.

Another consequence of [Lemma 1](#) is that, fixing μ^K , the designer's objective can be represented as maximizing a linear functional, where the choice variable is the slope of the indirect utility, a monotone function! Thus, the slope K for an extreme test allocation is a non-decreasing single-step function. By virtue of a linear objective and a single linear obedience constraint, an optimal allocation can be represented as a convex combination of at most two elements of $\text{ex}(\Lambda)$ (see [Dubins \(1960\)](#)). Thus, the optimal revenue is achieved by a mechanism where the slope of the indirect utility is a non-decreasing two-step function. In particular, the optimal test can be found among the ones with the following structure

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1, 1) & \mu \in [\mu_0, \mu_1) \\ (1, \alpha) & \mu \in [\mu_1, \bar{\mu}) \\ (1, 0) & \mu \in [\bar{\mu}, 1] \end{cases}$$

or

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1 - \alpha, 0) & \mu \in [\mu_0, \mu_1) \\ (1, \alpha) & \mu \in [\mu_1, \bar{\mu}) \\ (1, 0) & \mu \in [\bar{\mu}, 1] \end{cases}$$

The proposition then establishes sufficient conditions on the prior F under which 1) the optimal allocation is described by a single extreme point, 2) the optimal allocation is described by convex combination of two extreme points.

A.2 Proof of [Proposition 1](#)

Proof of [Proposition 1](#). First, we note that a test such that $q_0 = 0$ is never optimal; this can be seen as the obedience constraint is always slack for such tests.

To prove [item 1](#), we show by contradiction that a convex combination of two distinct tests can be improved. By [Lemma 1](#) we can restrict attention to tests where $q_1(\mu) = q_0(\mu)$ implies that either $q_1(\mu) = 0$ or $q_1(\mu) = 1$. First consider as a test (q_1, q_0) such that there exists $0 \leq \mu_0 \leq \mu_1 < \bar{\mu} \leq 1$ and

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1, 1) & \mu \in [\mu_0, \mu_1) \\ (1, \alpha) & \mu \in [\mu_1, \bar{\mu}) \\ (1, 0) & \mu \in [\bar{\mu}, 1] \end{cases}$$

We will construct a more profitable test $q' = (q'_1, q'_0)$ such that $q'_1 = q_1$ for all μ and $q'_0 = q_0$ on the complement of $[\mu_1, \bar{\mu})$. On $[\mu_1, \mu_1 + \varepsilon)$ set $q'_0 = 1$ for some small

$\varepsilon > 0$ and on $[\mu_1 + \varepsilon, \bar{\mu}]$ set $q'_0 = (1 - \delta)\alpha$ for small enough $\delta > 0$. Now we will argue that whenever $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is decreasing there exist $\varepsilon > 0$ and $\delta > 0$ such that q' is feasible and earns higher revenue. Revenue increases by

$$(1 - \alpha) \int_{\mu_1}^{\mu_1 + \varepsilon} \left[1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right] dF(\mu) - \delta \alpha \int_{\mu_1 + \varepsilon}^{\bar{\mu}} \left[1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right] dF(\mu)$$

and the slack in the obedience constraint increases by

$$-(1 - \alpha) \int_{\mu_1}^{\mu_1 + \varepsilon} \frac{(1 - \mu)p}{1 - p} dF(\mu) + \delta \alpha \int_{\mu_1 + \varepsilon}^{\bar{\mu}} \frac{(1 - \mu)p}{1 - p} dF(\mu)$$

For a fixed $\varepsilon > 0$ there exist a $\delta > 0$ such that both are positive if and only if

$$\frac{\int_{\mu_1}^{\mu_1 + \varepsilon} [1 - \mu + \frac{1 - F(\mu)}{f(\mu)}] dF(\mu)}{\int_{\mu_1 + \varepsilon}^{\bar{\mu}} [1 - \mu + \frac{1 - F(\mu)}{f(\mu)}] dF(\mu)} > \frac{\int_{\mu_1}^{\mu_1 + \varepsilon} \frac{(1 - \mu)p}{1 - p} dF(\mu)}{\int_{\mu_1 + \varepsilon}^{\bar{\mu}} \frac{(1 - \mu)p}{1 - p} dF(\mu)} = \frac{\int_{\mu_1}^{\mu_1 + \varepsilon} (1 - \mu) dF(\mu)}{\int_{\mu_1 + \varepsilon}^{\bar{\mu}} (1 - \mu) dF(\mu)}$$

and this inequality holds if and only if

$$\frac{\int_{\mu_1}^{\mu_1 + \varepsilon} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)}{\int_{\mu_1 + \varepsilon}^{\bar{\mu}} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)} > \frac{\int_{\mu_1}^{\mu_1 + \varepsilon} (1 - \mu) dF(\mu)}{\int_{\mu_1 + \varepsilon}^{\bar{\mu}} (1 - \mu) dF(\mu)} \quad (20)$$

For $\varepsilon = 0$ both sides are zero. Now suppose $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is decreasing in μ . Then

$$\frac{1 - F(\mu_1)}{(1 - \mu_1)f(\mu_1)} \geq \int_{\mu_1}^{\bar{\mu}} \frac{1 - F(\mu)}{(1 - \mu)f(\mu)} dF(\mu) \geq \frac{\int_{\mu_1}^{\bar{\mu}} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)}{\int_{\mu_1}^{\bar{\mu}} (1 - \mu) dF(\mu)}$$

or

$$\frac{\frac{1 - F(\mu_1)}{f(\mu_1)}}{\int_{\mu_1}^{\bar{\mu}} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)} \geq \frac{1 - \mu_1}{\int_{\mu_1}^{\bar{\mu}} (1 - \mu) dF(\mu)}$$

and the left-hand side is the derivative of the left-hand side in Equation 20 while the right-hand side is the derivative of the right-hand side in Equation 20, both at $\varepsilon = 0$. This guarantees that Equation 20 holds on a neighborhood of $\varepsilon = 0$.

Now consider some test (q_1, q_0) which has the following form

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1 - \alpha, 0) & \mu \in [\mu_0, \mu_1) \\ (1, \alpha) & \mu \in [\mu_1, \bar{\mu}) \\ (1, 0) & \mu \in [\bar{\mu}, 1) \end{cases}$$

If $\mu_0 = \mu_1$ and $\mu_1 < \bar{\mu}$ then previous argument shows that (q_1, q_0) is not optimal. When $\mu_0 < \mu_1$ and $\mu_1 < \bar{\mu}$, then we can use the same ideas to construct a profitable deviation, let (q'_1, q'_0) such that $q'_1 = q_1$ for all μ and $q_0 = q'_0$ on $([\mu_0, \bar{\mu})^C$. On $[\mu_0, \mu_0 + \varepsilon)$ set $q'_0 = 1 - \alpha$ for some small $\varepsilon > 0$ and on $[\mu_1, \bar{\mu})$ set $q'_0 = (1 - \delta)\alpha$ for small enough $\delta > 0$. Now we will argue that whenever $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is decreasing there exist $\varepsilon > 0$ and $\delta > 0$ such that the proposed deviation is feasible and leads to a higher revenue. The increase in revenue and obedience slack are

$$(1 - \alpha) \int_{\mu_0}^{\mu_0 + \varepsilon} [1 - \mu + \frac{1 - F(\mu)}{f(\mu)}] dF(\mu) - \delta \alpha \int_{\mu_1}^{\bar{\mu}} [1 - \mu + \frac{1 - F(\mu)}{f(\mu)}] dF(\mu)$$

and

$$-(1 - \alpha) \int_{\mu_0}^{\mu_0 + \varepsilon} \frac{(1 - \mu)p}{1 - p} dF(\mu) + \delta \alpha \int_{\mu_1}^{\bar{\mu}} \frac{(1 - \mu)p}{1 - p} dF(\mu),$$

respectively. For a fixed $\varepsilon > 0$ there exist a $\delta > 0$ such that both are positive if and only if

$$\frac{\int_{\mu_0}^{\mu_0 + \varepsilon} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)}{\int_{\mu_1}^{\bar{\mu}} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)} > \frac{\int_{\mu_0}^{\mu_0 + \varepsilon} 1 - \mu dF(\mu)}{\int_{\mu_1}^{\bar{\mu}} 1 - \mu dF(\mu)}$$

which can be shown by a similar derivation as above to be true for small enough $\varepsilon > 0$. We have established [item 1](#) that $(q_1, q_0) \in \text{ex}(\Lambda)$.

To prove [item 2](#), consider some $(q_1, q_0) \in \text{ex}(\Lambda)$ and $0 \leq \mu_0 < \bar{\mu} < 1$ such that

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1, 1) & \mu \in [\mu_0, \bar{\mu}) \\ (1, 0) & \mu \in [\bar{\mu}, 1] \end{cases}$$

We will show that there exist tests (q'_1, q'_0) and (q''_1, q''_0) such that a mixture of these is feasible and more profitable than $(q_1(\mu), q_0(\mu))$. To this end for some $\varepsilon > 0$ define

$$(q'_1(\mu), q'_0(\mu)) := \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1, 1) & \mu \in [\mu_0, \bar{\mu} - \varepsilon) \\ (1, 0) & \mu \in [\bar{\mu} - \varepsilon, 1] \end{cases}$$

$$\text{and } (q''_1(\mu), q''_0(\mu)) := \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1, 1) & \mu \in [\mu_0, \bar{\mu} + \varepsilon) \\ (1, 0) & \mu \in [\bar{\mu} + \varepsilon, 1] \end{cases}$$

Define revenue and obedience functions on $[\mu_0, 1]$:

$$R(\mu) := \int_{\mu_0}^{\mu} dF(\tilde{\mu}) + \int_{\mu}^1 [\tilde{\mu} - \frac{1 - F(\tilde{\mu})}{f(\tilde{\mu})}] dF(\tilde{\mu})$$

$$Q(\mu) := \int_{\mu_0}^{\mu} \frac{\tilde{\mu} - p}{1 - p} dF(\tilde{\mu}) + \int_{\mu}^1 \tilde{\mu} dF(\tilde{\mu})$$

Note that R and $-Q$ are differentiable and increasing. Let

$$\alpha' := \inf\{\alpha \in [0, 1] | \alpha R(\bar{\mu} - \varepsilon) + (1 - \alpha)R(\bar{\mu} + \varepsilon) = R(\bar{\mu})\}$$

$$\alpha'' := \inf\{\alpha \in [0, 1] | \alpha Q(\bar{\mu} - \varepsilon) + (1 - \alpha)Q(\bar{\mu} + \varepsilon) = Q(\bar{\mu})\}$$

Also define the corresponding types $\mu' := \bar{\mu} + (1 - 2\alpha')\varepsilon$ and $\mu'' := \bar{\mu} + (1 - 2\alpha'')\varepsilon$. In particular, we get that

$$\alpha' = \frac{R(\bar{\mu} + \varepsilon) - R(\bar{\mu})}{R(\bar{\mu} + \varepsilon) - R(\bar{\mu} - \varepsilon)} \quad \text{and} \quad \alpha'' = \frac{Q(\bar{\mu}) - Q(\bar{\mu} + \varepsilon)}{Q(\bar{\mu} - \varepsilon) - Q(\bar{\mu} + \varepsilon)}$$

The proof follows by showing that $\mu' < \mu''$, for which it is sufficient to show $\alpha' > \alpha''$.

$$\begin{aligned} & \frac{R(\bar{\mu} + \varepsilon) - R(\bar{\mu})}{R(\bar{\mu} + \varepsilon) - R(\bar{\mu} - \varepsilon)} > \frac{Q(\bar{\mu}) - Q(\bar{\mu} + \varepsilon)}{Q(\bar{\mu} - \varepsilon) - Q(\bar{\mu} + \varepsilon)} \\ \iff & \frac{\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon} [1 - \mu + \frac{1 - F(\mu)}{f(\mu)}] dF(\mu)}{\int_{\bar{\mu} - \varepsilon}^{\bar{\mu} + \varepsilon} [1 - \mu + \frac{1 - F(\mu)}{f(\mu)}] dF(\mu)} > \frac{\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon} (1 - \mu) dF(\mu)}{\int_{\bar{\mu} - \varepsilon}^{\bar{\mu} + \varepsilon} (1 - \mu) dF(\mu)} \\ \iff & \frac{\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)}{\int_{\bar{\mu} - \varepsilon}^{\bar{\mu} + \varepsilon} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)} > \frac{\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon} (1 - \mu) dF(\mu)}{\int_{\bar{\mu} - \varepsilon}^{\bar{\mu} + \varepsilon} (1 - \mu) dF(\mu)} \\ \iff & \frac{\int_{\bar{\mu} - \varepsilon}^{\bar{\mu}} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)}{\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)} < \frac{\int_{\bar{\mu} - \varepsilon}^{\bar{\mu}} (1 - \mu) dF(\mu)}{\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon} (1 - \mu) dF(\mu)} \end{aligned}$$

Whenever $\frac{1 - F(\mu)}{(1 - \mu)f(\mu)}$ is non-decreasing the required inequality holds as

$$\frac{\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon} \frac{1 - F(\mu)}{f(\mu)} dF(\mu)}{\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon} (1 - \mu) dF(\mu)} \geq \frac{1 - F(\bar{\mu})}{(1 - \bar{\mu})f(\bar{\mu})} \geq \frac{\int_{\bar{\mu} - \varepsilon}^{\bar{\mu}} (\frac{1 - F(\mu)}{f(\mu)}) dF(\mu)}{\int_{\bar{\mu} - \varepsilon}^{\bar{\mu}} (1 - \mu) dF(\mu)}$$

Recall that any test can be written as a convex combination of at most two extreme points. The above argument shows that if $\frac{1 - F(\mu)}{(1 - \mu)f(\mu)}$ is non-decreasing, then the optimal solution is represented by a convex combination of extreme points for which

either $\bar{\mu} = 1$ or $\bar{\mu} = \mu_0$. Using [Lemma 1](#), the optimal test can be expressed in the following form:

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1 - \alpha, 0) & \mu \in [\mu_0, \bar{\mu}) \\ (1, \alpha) & \mu \in [\bar{\mu}, 1] \end{cases}$$

□

B Proof of Theorem 1

Proof of Theorem 1. Fix a mechanism maximizing [Equation 3](#), subject to the envelope representation in [Section 3](#) and the obedience, Bayes-plausibility, and participation requirements in [Section 2](#). Thus $U(\mu) \geq U^*$ for every type, while the monotonicity constraint in [Equation 4](#) is relaxed. We establish this constraint by contradiction.

Write $\rho_\theta(\mu)$ for the action marginal of the test. In the sender-linear environment of [Section 4](#), define the state-contingent payoffs and use the slope in [Equation 4](#):

$$q_\theta(\mu) = \int_A \pi_S(a, \theta) d\rho_\theta(\mu)(a), \quad K(\mu) = q_1(\mu) - q_0(\mu). \quad (21)$$

The compactness of A and continuity of $\pi_S(\cdot, \theta)$ assumed in [Theorem 1](#) bound these payoffs and hence K . The envelope representation therefore makes U continuous, so its minimum is attained. At a revenue maximum this minimum equals U^* , since otherwise all fees could be increased by a common amount. Let μ^0 be the smallest type whose participation constraint binds. Anchoring the envelope representation in [Section 3](#) at μ^0 and changing the order of integration in the objective [Equation 3](#) gives

$$\Pi = \mathbb{E}v - U^* + \int_0^{\mu^0} F(t)K(t) dt - \int_{\mu^0}^1 [1 - F(t)]K(t) dt. \quad (22)$$

The envelope contribution below μ^0 involves F , and that above μ^0 involves $1 - F$. The exchange argument uses the corresponding distributional condition on each side.

Exchanging tests. Suppose, for contradiction, that an interval where participation is strict contains measurable sets S_1 and S_2 of positive F -measure such that

$$\sup S_1 < \inf S_2, \quad \text{ess inf}_{S_1} K > \text{ess sup}_{S_2} K. \quad (23)$$

Choose these sets inside a closed subinterval where $U > U^*$. Since $K = q_1 - q_0$ by [Equation 21](#), restricting the sets further if necessary gives at least one of

$$\operatorname{ess\,inf}_{S_1} q_1 > \operatorname{ess\,sup}_{S_2} q_1, \quad \theta = 1, \quad (24)$$

$$\operatorname{ess\,sup}_{S_1} q_0 < \operatorname{ess\,inf}_{S_2} q_0, \quad \theta = 0. \quad (25)$$

Define the state-weighted measures by

$$d\gamma_1(\mu) = \mu dF(\mu), \quad d\gamma_0(\mu) = (1 - \mu) dF(\mu). \quad (26)$$

Use state $\theta = 1$ in [Equation 24](#) and state $\theta = 0$ in [Equation 25](#). Continuity of F allows us to reduce one of the sets so that $\gamma_\theta(S_1) = \gamma_\theta(S_2) > 0$. For $i = 1, 2$, the conditional distribution on S_i is

$$\gamma_\theta^i(B) = \frac{\gamma_\theta(B \cap S_i)}{\gamma_\theta(S_i)}. \quad (27)$$

Let $T_\theta^2 : S_1 \rightarrow S_2$ and $T_\theta^1 : S_2 \rightarrow S_1$ send each type to the same quantile of the other conditional distribution. These monotone maps satisfy ([Santambrogio, 2015](#), Theorem 2.5)

$$(T_\theta^j)_\# \gamma_\theta^i = \gamma_\theta^j, \quad i \neq j. \quad (28)$$

To obtain a profitable perturbation, exchange a fraction $\varepsilon \in (0, 1)$ of each test with that of its matched type:

$$\rho'_\theta(\mu) = \begin{cases} (1 - \varepsilon)\rho_\theta(\mu) + \varepsilon\rho_\theta(T_\theta^2(\mu)), & \mu \in S_1, \\ (1 - \varepsilon)\rho_\theta(\mu) + \varepsilon\rho_\theta(T_\theta^1(\mu)), & \mu \in S_2, \\ \rho_\theta(\mu), & \text{otherwise.} \end{cases} \quad (29)$$

Leave the other state's test unchanged and disclose the posterior generated by the perturbed test and realized action. Equal state-weighted masses and [Equation 28](#) preserve the joint distribution of actions and the state. Receiver linearity therefore preserves obedience, and [Equation 21](#) preserves expected gross surplus.

Let q'_θ be the payoffs induced by [Equation 29](#) through [Equation 21](#), and let $K'(\mu) = q'_1(\mu) - q'_0(\mu)$ be its envelope slope. In either [Equation 24](#) or [Equation 25](#), $\Delta K(\mu) := K'(\mu) - K(\mu)$ is negative on S_1 , positive on S_2 , and zero elsewhere. Moreover, [Equation 28](#) gives

$$\int_0^1 \Delta K(t) d\gamma_\theta(t) = 0. \quad (30)$$

Participation and revenue. Although μ^0 is the smallest binding type, participation may also bind at larger types. Preserving expected gross surplus alone does not

preserve their envelope utilities. Define the utility and fees of the perturbed test by

$$\begin{aligned} U'(\mu) &= U(\mu) + \max \left\{ 0, - \int_0^1 \Delta K(t) dt \right\} + \int_0^\mu \Delta K(t) dt, \\ \phi'(\mu) &= v'(\mu) - U'(\mu). \end{aligned} \quad (31)$$

Below both S_1 and S_2 , the utility change is the nonnegative constant in Equation 31; above both sets it equals $\max\{0, \int_0^1 \Delta K(t) dt\} \geq 0$. Participation is therefore preserved outside the closed interval containing the exchange. Inside it, continuity of U gives a strictly positive minimum slack, while the utility change tends uniformly to zero with ε . A sufficiently small perturbation thus preserves participation for every type.

The perturbed fees satisfy the envelope representation. Since expected gross surplus is unchanged, applying the objective Equation 3 to Equation 31 yields

$$\begin{aligned} \Delta \Pi &= - \max \left\{ 0, - \int_0^1 \Delta K(t) dt \right\} - \int_0^1 [1 - F(t)] \Delta K(t) dt \\ &= \min \left\{ - \int_0^1 [1 - F(t)] \Delta K(t) dt, \int_0^1 F(t) \Delta K(t) dt \right\}. \end{aligned} \quad (32)$$

If $\int_0^1 \Delta K(t) dt \geq 0$, the constant adjustment is zero and the first expression in the minimum applies. Otherwise, the adjustment adds $\int_0^1 \Delta K(t) dt$ to the revenue change, giving $\int_0^1 F(t) \Delta K(t) dt$. This proves the second line of Equation 32. It remains to show that the revenue change is strictly positive in both cases.

By Equation 26, $d\gamma_1(t) = tf(t) dt$ and $d\gamma_0(t) = (1 - t)f(t) dt$. Dividing the coefficient by the corresponding density rewrites the first expression as

$$- \int_0^1 [1 - F(t)] \Delta K(t) dt = \begin{cases} - \int_0^1 \frac{1 - F(t)}{tf(t)} \Delta K(t) d\gamma_1(t), & \theta = 1, \\ - \int_0^1 \frac{1 - F(t)}{(1 - t)f(t)} \Delta K(t) d\gamma_0(t), & \theta = 0. \end{cases} \quad (33)$$

The coefficient $\frac{1 - F(t)}{(1 - t)f(t)}$ is strictly decreasing by Theorem 1, and so is $\frac{1 - F(t)}{tf(t)} = \frac{1 - t}{t} \frac{1 - F(t)}{(1 - t)f(t)}$. In each line, subtract the coefficient's value at a point between S_1 and S_2 . The integral is unchanged by Equation 30; the product with ΔK is then negative on both S_1 and S_2 . The minus sign in Equation 33 therefore makes this revenue change strictly positive.

For the second expression, the same substitution gives

$$\int_0^1 F(t) \Delta K(t) dt = \begin{cases} \int_0^1 \frac{F(t)}{tf(t)} \Delta K(t) d\gamma_1(t), & \theta = 1, \\ \int_0^1 \frac{F(t)}{(1-t)f(t)} \Delta K(t) d\gamma_0(t), & \theta = 0. \end{cases} \quad (34)$$

Here $\frac{F(t)}{tf(t)}$ is strictly increasing by [Theorem 1](#), and so is $\frac{F(t)}{(1-t)f(t)} = \frac{t}{1-t} \frac{F(t)}{tf(t)}$. Subtracting the coefficient's value between S_1 and S_2 now makes the product positive on both sets. Thus [Equation 34](#) is strictly positive as well. Only $S_1 \cup S_2$ contributes to these integrals, so all terms are finite. The strictly positive revenue change in [Equation 32](#) contradicts optimality.

Consequently, K is non-decreasing almost everywhere on each interval where participation is strict, and U is convex there. Two binding types cannot enclose an interval of strict participation: convexity would put utility weakly below U^* between them. The binding set is therefore an interval, possibly a single point or all of $[0, 1]$. The slope is non-positive below μ^0 , zero on the binding interval, and non-negative thereafter, almost everywhere. Hence U is globally convex and its slope satisfies [Equation 4](#) up to a set of F -measure zero.

We can therefore make K non-decreasing everywhere by adjusting the tests and fees on a set of F -measure zero, yielding an incentive-compatible mechanism with unchanged revenue. $\square$

As an aside note that the two conditions on the CDF are mutually consistent. For example, $f(\mu) = 6\mu(1 - \mu)$ and $F(\mu) = 3\mu^2 - 2\mu^3$ give

$$\frac{1 - F(\mu)}{(1 - \mu)f(\mu)} = \frac{1}{3} + \frac{1}{6\mu}, \quad \frac{F(\mu)}{\mu f(\mu)} = \frac{1}{3} + \frac{1}{6(1 - \mu)}.$$

Both required monotonicities are strict on $(0, 1)$; the density may vanish at the endpoints.

C Proof of [Theorem 2](#) and [Corollary 1](#)

Proof. Condition on an equilibrium price recommendation $p \in (0, 1)$ and platform access, and let G be the resulting obedient belief distribution, supported on $[p, 1]$. The argument applies to every recommendation with positive probability and almost surely for a continuously distributed recommendation. We first find a distribution H_x^p with the same mean that is more dispersed in convex order.

Feasible mean. The mean of an obedient belief distribution cannot exceed $\mathbf{E}H_0^p = p(1 - \ln p)$. To see this, suppose $\mathbf{E}G > \mathbf{E}H_0^p$. Since both distributions are supported

on $[p, 1]$, comparison of their integrated CDFs gives some $z \in (p, 1)$ with $G(z) < H_0^p(z) = 1 - \frac{p}{z}$. The seller could then increase revenue by charging z :

$$z[1 - G(z-)] \geq z[1 - G(z)] > p,$$

contradicting obedience. Here $G(z-) = \lim_{y \uparrow z} G(y)$, so the demand includes buyers whose belief equals the price. We conclude that

$$p \leq \mathbf{E}G \leq \mathbf{E}H_0^p. \quad (35)$$

More generally, obedience gives $1 - G(z-) \leq \frac{p}{z}$ for every $p < z \leq 1$, including any atom at one. In particular, $G(z) \geq H_0^p(z)$ for $p \leq z < 1$.

From Equation 12,

$$\mathbf{E}H_x^p = p + \int_p^1 \min\{1 - x, p/z\} dz.$$

This is continuous and strictly decreasing in x on $[0, 1]$, with the displayed endpoint means. Strictness follows because increasing x strictly lowers the integrand on an interval immediately above p . There is therefore a unique x for which $\mathbf{E}H_x^p = \mathbf{E}G$.

Put $t = \min\{p/(1 - x), 1\}$, with $t = 1$ when $x = 1$, and define

$$D(z) = \int_p^z [H_x^p(s) - G(s)] ds.$$

Equal means give $D(1) = 0$. On $[t, 1)$, $H_x^p = H_0^p \leq G$, so $D(z) = -\int_z^1 [H_x^p(s) - G(s)] ds \geq 0$ for $z \geq t$. On $[p, t]$, D is concave because its almost-everywhere derivative is $x - G(z)$, a non-increasing function. Since $D(p) = 0$ and $D(t) \geq 0$, it follows that $D \geq 0$ throughout this interval too. This includes $t = 1$ and the collapsed interval $t = p$. The integrated CDF characterization of convex order now gives $H_x^p \succeq_{\text{mps}} G$. Endpoint atoms are included through the CDFs and the equality of means.

By the martingale characterization of convex order (Strassen, 1965), the spread from G to H_x^p can be implemented by additional private information after the allocation and price recommendation. Conditional means are preserved, so Bayes' rule implements this information without changing the state-contingent allocation of platform access or prices for any report. Fees are unchanged. Under the enforcement convention, report payoffs depend only on that allocation and the fees, hence revenue, report IC, and interim participation are preserved. Truthful buyers remain willing to purchase because their resulting beliefs lie in $[p, 1]$. Finally, $z\text{Prob}(v \geq z) \leq p$ under H_x^p for every z , including the atom at one, so the seller remains obedient.

Apply the construction conditional on each recommended price, preserving the state-contingent allocation and fees for every report. Recommendations of zero or one already induce degenerate beliefs and require no change. This proves [Theorem 2](#).

For [Corollary 1](#), the same construction applied directly to G_q proves sufficiency. Conversely, any obedient information policy implementing the allocation induces a belief law $G \succeq_{\text{mps}} G_q$, by conditional Jensen's inequality. The preceding argument gives $H_x^p \succeq_{\text{mps}} G$ for its mean, and transitivity proves necessity. No restriction on the prior beyond the maintained model is needed for this obedience characterization. $\square$

D Digital Platform

D.1 Relaxed Problem

Recall that $K(\mu) = (1 - p)q_1(\mu) + pq_0(\mu)$ is the slope of the indirect utility. We may substitute into the objective in [Equation 16](#) to obtain

$$\begin{aligned} \Pi = \int_0^1 \mu(1 - p)q_1(\mu)dF(\mu) - \int_0^1 p(1 - \mu)q_0(\mu)dF(\mu) \\ - \int_0^1 K(\mu) \left(\frac{1 - F(\mu)}{f(\mu)} \right) dF(\mu) \end{aligned}$$

For any incentive-compatible and obedient mechanism q there is a corresponding monotone slope function $K(\mu)$ taking values in $[0, 1]$ and a target *platform mean* m defined by

$$\int_0^1 \mu q_1(\mu)dF(\mu) = \frac{m}{1 - m} \int_0^1 (1 - \mu)q_0(\mu)dF(\mu)$$

or

$$\int_0^1 \left[1 - p - \mu \left(1 - \frac{p}{m} \right) \right] q_1(\mu)dF(\mu) = \int_0^1 (1 - \mu)K(\mu)dF(\mu) \quad (36)$$

Substituting again into the objective and re-arranging we arrive at:

$$\Pi = \int_0^1 \mu \left(1 - \frac{p}{m} \right) q_1(\mu)dF(\mu) - \int_0^1 \frac{1 - F(\mu)}{f(\mu)} K(\mu)dF(\mu) \quad (37)$$

We have organized the objective function in a way that isolates the gains from increasing $q_1(\mu)$ from the costs of increasing the slope $K(\mu)$. In the background

we can adjust $q_0(\mu)$ to maintain a given slope $K(\mu)$ and target mean m provided $q_1(\mu) \in [0, 1]$ satisfies

$$\frac{K(\mu) - p}{1 - p} \leq q_1(\mu) \leq \frac{K(\mu)}{1 - p}. \quad (38)$$

The first inequality ensures $q_0(\mu) \leq 1$ and the second ensures $q_0(\mu) \geq 0$. The participation constraint [Equation 6](#) requires

$$q_1(\mu) \geq \frac{(1 - \mu)K(\mu)}{1 - p}. \quad (39)$$

We will say that a slope function $K(\mu)$ is *feasible* with respect to a target mean m if there exists an allocation $q_1(\mu)$ which satisfies [Equation 36](#), [Equation 38](#), and [Equation 39](#).

D.2 Optimal Allocation for Fixed K

We can approach the problem by first finding the optimal q_1 for a given target mean m and feasible $K(\mu)$, and then optimizing the latter.

Define the following class of bang-bang allocations.

$$q_1^K(\mu) := \begin{cases} \max \left\{ \frac{(1-\mu)K(\mu)}{1-p}, \frac{K(\mu)-p}{1-p} \right\} & \mu < \mu^K \\ \min \left\{ 1, \frac{K(\mu)}{1-p} \right\} & \mu \geq \mu^K \end{cases}$$

For some $\mu^K \in [0, 1]$.

Lemma 2. *For given m and feasible $K(\mu)$, the optimal allocation is $q_1 = q_1^K$ for some μ^K .*

Proof of Lemma 2. Consider any q_1 for which there is an interval of types, $[\mu_0, \bar{\mu}]$ such that $\max \left\{ \frac{(1-\mu)K(\mu)}{1-p}, \frac{K(\mu)-p}{1-p} \right\} < q_1(\mu) < \min \left\{ 1, \frac{K(\mu)}{1-p} \right\}$. Pick $\delta > 0$ such that $\bar{\mu} - \mu_0 \geq 2\delta$. We can improve the revenue by increasing q_1 on $[\bar{\mu} - \delta, \bar{\mu}]$ by some $\varepsilon_1 > 0$ and by reducing q_1 on $[\mu_0, \mu_0 + \delta]$ by $\varepsilon_0 > 0$ where

$$\varepsilon_1 = \varepsilon_0 \frac{\int_{\mu_0}^{\mu_0 + \delta} (1 - p - \mu (1 - \frac{p}{m})) dF(\mu)}{\int_{\bar{\mu} - \delta}^{\bar{\mu}} (1 - p - \mu (1 - \frac{p}{m})) dF(\mu)}$$

ensuring that the target mean, i.e. [Equation 36](#) is maintained. This adjustment doesn't violate the constraint in [Equation 15](#) as it weakly decreases the size of any point mass at p without changing the target mean.

The revenue from the new allocation is greater than the old allocation if the following holds:

$$\begin{aligned}
\frac{\int_{\mu_0}^{\mu_0+\delta} (1-p-\mu(1-\frac{p}{m})) dF(\mu)}{\int_{\bar{\mu}-\delta}^{\bar{\mu}} (1-p-\mu(1-\frac{p}{m})) dF(\mu)} &\geq \frac{\int_{\mu_0}^{\mu_0+\delta} \mu(1-\frac{p}{m}) dF(\mu)}{\int_{\bar{\mu}-\delta}^{\bar{\mu}} \mu(1-\frac{p}{m}) dF(\mu)} \\
\iff \frac{\int_{\mu_0}^{\mu_0+\delta} (1-p) dF(\mu)}{\int_{\bar{\mu}-\delta}^{\bar{\mu}} (1-p) dF(\mu)} &\geq \frac{\int_{\mu_0}^{\mu_0+\delta} \mu(1-\frac{p}{m}) dF(\mu)}{\int_{\bar{\mu}-\delta}^{\bar{\mu}} \mu(1-\frac{p}{m}) dF(\mu)} \\
\iff \frac{\int_{\mu_0}^{\mu_0+\delta} dF(\mu)}{\int_{\bar{\mu}-\delta}^{\bar{\mu}} dF(\mu)} &\geq \frac{\int_{\mu_0}^{\mu_0+\delta} \mu dF(\mu)}{\int_{\bar{\mu}-\delta}^{\bar{\mu}} \mu dF(\mu)}
\end{aligned}$$

The last inequality follows from our choice of δ .

Now consider q_1 such that there are intervals $I_1 < I_2$ where

$$q_1(\mu) = \max \left\{ \frac{(1-\mu)K(\mu)}{1-p}, \frac{K(\mu)-p}{1-p} \right\}$$

for $\mu \in I_2$ and

$$q_1(\mu) = \min \left\{ 1, \frac{K(\mu)}{1-p} \right\}$$

for $\mu \in I_1$. We construct an improvement similar to the above by slightly increasing q_1 on I_2 while reducing q_1 on I_1 to keep the mean constraint binding. Finally, note that this improvement introduces more slack to [Equation 15](#). The revenue of this improvement is greater than the old allocation if the following holds:

$$\frac{\int_{I_1} dF(\mu)}{\int_{I_2} dF(\mu)} \geq \frac{\int_{I_1} \mu dF(\mu)}{\int_{I_2} \mu dF(\mu)}$$

This is implied by $\frac{\inf(I_2)}{\sup(I_1)} \geq 1$. □

D.3 Optimal Choice of K

Notice that the monotonicity of $K(\mu)$ implies that there exists μ_0 such that

$$\frac{(1-\mu)K(\mu)}{1-p} \geq \frac{K(\mu)-p}{1-p}$$

for all $\mu \leq \mu_0$ and

$$\frac{(1-\mu)K(\mu)}{1-p} < \frac{K(\mu)-p}{1-p}$$

for all $\mu > \mu_0$. In particular the participation constraint in Equation 6 binds only for types $\mu \leq \mu_0$ and all mass x at the point p comes from these types. The next lemma states that $x = 0$ for a solution to the relaxed problem.

Lemma 3. *The solution to the relaxed problem has $K(\mu) = 0$ for all $\mu < \mu_0$.*

Proof of Lemma 3. Consider a feasible allocation with target mean m and mass size x . Suppose there is a non-empty interval $[\underline{\mu}, \mu_0)$ consisting of types μ for which $q_1(\mu) = \frac{(1-\mu)K(\mu)}{1-p} > 0$. Then $x > 0$ and $m \leq EH_0^p$.

Consider a new allocation which reduces $K(\mu)$ slightly at all points in $[\underline{\mu}, \mu_0)$ while keeping q_1 unchanged. The new allocation increases the mean slightly (by Equation 14) to say $m' > m$ and eliminates any mass point (Equation 6 will be slack). For m' close enough to m the new allocation is feasible for the relaxed problem with target mean m' and has slack Equation 6. That is, Equation 14 and Equation 15 are satisfied for m' and $x = 0$ because $EH_0^p > EH_x^p$.

By inspection of the objective in Equation 37 this is an improvement. Therefore a solution to the relaxed problem must have $K(\mu) = 0$ for all $\mu < \mu_0$. □

Lemma 2 and Lemma 3 imply that a solution to the relaxed problem can be described by μ_0 and two further thresholds $\mu_0 \leq \mu_1 \leq \bar{\mu}$ such that

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu < \mu_0 \\ \left(\frac{K(\mu)-p}{1-p}, 1\right) & \mu \in [\mu_0, \mu_1) \\ \left(\frac{K(\mu)}{1-p}, 0\right) & \mu \in [\mu_1, \bar{\mu}) \\ \left(1, \frac{1}{p}(K(\mu) - (1-p))\right) & \mu > \bar{\mu} \end{cases} \quad (40)$$

Lemma 4. *A solution to the relaxed problem has a threshold $\mu' \geq \bar{\mu}$ such that*

1. $K(\mu) = 1$ for all $\mu \geq \mu'$.
2. $q_0(\mu)$ is constant on $[\mu_0, \mu')$ and equal to 0 or 1.

Proof of Lemma 4. We first show that any candidate solution can be weakly improved by one that has $q_0(\mu) \in \{0, 1\}$ for all $\mu \in [\bar{\mu}, 1]$.⁵ If the candidate allocation

⁵By Equation 40, $K(\mu) \leq 1 - p$ on $[\mu_1, \bar{\mu})$, whereas $K(\mu) = 1 - p + pq_0(\mu) \geq 1 - p$ above $\bar{\mu}$. Thus reducing q_0 at the start of the upper region preserves the boundary ordering. If $\mu_1 = \bar{\mu}$, instead begin the reduction just above μ_0 , below which the slope is zero. A small constant reduction preserves monotonicity within this interval, creates an upward jump at its right endpoint, and relaxes purchase participation.

does not already have that property then we construct a new allocation q' such that on $q'(\mu) = q(\mu)$ for $\mu \in [\bar{\mu}, 1]^C$. On the interval $[\bar{\mu}, \bar{\mu} + \varepsilon_0)$ let $q'_0(\mu) = q_0(\bar{\mu})$, on the interval $[1 - \varepsilon_1, 1]$ let $q'_0(\mu) = 1$, and $q'_0(\mu) = q_0(\mu)$ for $\mu \in [\bar{\mu} + \varepsilon_0, 1 - \varepsilon_1)$. Where

$$\int_{\bar{\mu}}^{\bar{\mu} + \varepsilon_0} (1 - \mu)(q_0(\mu) - q_0(\bar{\mu}))dF(\mu) - \int_{1 - \varepsilon_1}^1 (1 - \mu)(1 - q_0(\mu))dF(\mu) = 0$$

As long as $\bar{\mu} + \varepsilon_0 \leq 1 - \varepsilon_1$, the allocation q' is well-defined and has the same target mean as q . Consider the difference in revenue between q and q' :

$$\begin{aligned} & \int_{\bar{\mu}}^{\bar{\mu} + \varepsilon_0} \left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)}\right) (q_0(\mu) - q_0(\bar{\mu}))dF(\mu) \\ & \quad - \int_{1 - \varepsilon_1}^1 \left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)}\right) (1 - q_0(\mu))dF(\mu) \\ &= \int_{\bar{\mu}}^{\bar{\mu} + \varepsilon_0} \frac{1 - F(\mu)}{f(\mu)} (q_0(\mu) - q_0(\bar{\mu}))dF(\mu) - \int_{1 - \varepsilon_1}^1 \frac{1 - F(\mu)}{f(\mu)} (1 - q_0(\mu))dF(\mu) \end{aligned}$$

When $\frac{1 - F(\mu)}{(1 - \mu)f(\mu)}$ is non-increasing the change in revenue is positive as

$$\begin{aligned} & \int_{1 - \varepsilon_1}^1 \frac{1 - F(\mu)}{f(\mu)} (1 - q_0(\mu))dF(\mu) \leq \frac{1 - F(1 - \varepsilon_1)}{\varepsilon_1 f(1 - \varepsilon_1)} \int_{1 - \varepsilon_1}^1 (1 - \mu)(1 - q_0(\mu))dF(\mu) \\ & \implies \int_{1 - \varepsilon_1}^1 \frac{1 - F(\mu)}{f(\mu)} (1 - q_0(\mu))dF(\mu) \\ & \quad \leq \frac{1 - F(1 - \varepsilon_1)}{\varepsilon_1 f(1 - \varepsilon_1)} \int_{\bar{\mu}}^{\bar{\mu} + \varepsilon_0} (1 - \mu)(q_0(\mu) - q_0(\bar{\mu}))dF(\mu) \\ & \implies \int_{1 - \varepsilon_1}^1 \frac{1 - F(\mu)}{f(\mu)} (1 - q_0(\mu))dF(\mu) \leq \int_{\bar{\mu}}^{\bar{\mu} + \varepsilon_0} \frac{1 - F(\mu)}{f(\mu)} (q_0(\mu) - q_0(\bar{\mu}))dF(\mu) \end{aligned}$$

The first implication follows from the choice of $\varepsilon_0, \varepsilon_1$. Thus any candidate allocation can be weakly improved by one for which there exists $\mu' \in [\bar{\mu}, 1]$ such that $q_0(\mu) = 0$ on $[\bar{\mu}, \mu')$ and $q_0(\mu) = 1$ on $[\mu', 1]$. Note that the latter implies [item 1](#) in the statement of the Lemma in view of [Equation 40](#).

In particular, by [Equation 40](#), the improved allocation has $q_0(\mu) = 1$ for $\mu \in [\mu_0, \mu_1)$ and $q_0(\mu) = 0$ for $\mu \in [\mu_1, \mu')$. Next we claim that any such allocation for which μ_1 is strictly between μ_0 and μ' can be improved by one which satisfies [item 2](#) in the statement of the Lemma. In other words, one for which μ_1 equals either μ_0 or μ' .

Assume for contradiction that the q is such that $\mu_0 < \mu_1 < \mu'$. We construct a new allocation q' such that on $q'(\mu) = q(\mu)$ for $\mu \in [\mu_0, \mu')^C$. On the interval $[\mu_0, \mu_1)$ we

require $q'_0(\mu) = q_0(\mu) - \varepsilon_0$, on the interval $[\mu_1, \mu']$ we require $q'_0(\mu) = q_0(\mu) + \varepsilon_1$. Where

$$\varepsilon_1 = \varepsilon_0 \frac{\int_{\mu_0}^{\mu_1} (1 - \mu) dF(\mu)}{\int_{\mu_1}^{\mu'} (1 - \mu) dF(\mu)}$$

For small enough $\varepsilon_0 > 0$, the allocation q' is feasible with the same target mean as q . We argue that under the condition of the proposition, this q' achieves a weakly higher revenue. To see this, consider the difference in revenue between q and q' :

$$\varepsilon_0 \int_{\mu_0}^{\mu_1} \left[1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right] dF(\mu) - \varepsilon_1 \int_{\mu_1}^{\mu'} \left[1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right] dF(\mu)$$

Similar to [Proposition 1](#), when $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-increasing the change in revenue is positive as

$$\frac{\int_{\mu_0}^{\mu_1} \frac{1-F(\mu)}{f(\mu)} dF(\mu)}{\int_{\mu_1}^{\mu'} \frac{1-F(\mu)}{f(\mu)} dF(\mu)} > \frac{\int_{\mu_0}^{\mu_1} (1 - \mu) dF(\mu)}{\int_{\mu_1}^{\mu'} (1 - \mu) dF(\mu)}.$$

□

D.4 Proof of [Proposition 2](#)

Proof of [Proposition 2](#): It follows from [Lemma 3](#), [Lemma 4](#) and [Equation 40](#) that a solution to the relaxed problem takes one of the following two forms.

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu < \mu_0 \\ \left(\frac{K(\mu) - p}{1 - p}, 1 \right) & \mu \in [\mu_0, \mu'] \\ (1, 1) & \mu > \mu' \end{cases} \quad (41)$$

or

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0) & \mu < \mu_0 \\ \left(\frac{K(\mu)}{1 - p}, 0 \right) & \mu \in [\mu_0, \mu'] \\ (1, 1) & \mu > \mu' \end{cases} \quad (42)$$

We complete the proof of [Proposition 2](#) by identifying the slope in the middle regions. Consider the first case and let $K(\mu)$ be a feasible slope. By definition of μ_0 we have $K(\mu_0) - p \geq (1 - \mu_0)K(\mu_0)$ implying $K(\mu_0) \geq \frac{p}{\mu_0}$. Therefore by monotonicity of K , for all $\mu \geq \mu_0$ we have $1 \geq K(\mu) \geq \frac{p}{\mu_0}$. We will show in fact that a solution to the relaxed problem in this case has $K(\mu) \in \left\{ \frac{p}{\mu_0}, 1 \right\}$ for all $\mu \in [\mu_0, \mu']$.

If $K(\mu)$ fails this property then we construct a new allocation that is determined by Equation 41 for slope function $\hat{K}$, where $\hat{K}(\mu) = K(\mu)$ for all $\mu \notin [\mu_0, \mu']$. Let $\hat{K}(\mu) = \frac{p}{\mu_0}$ for $\mu \in [\mu_0, \mu_0 + \varepsilon_0]$ and $\hat{K}(\mu) = 1$ for $\mu \in [\mu' - \varepsilon_1, \mu']$ and $\hat{K} = K$ otherwise. Choose $\varepsilon_0, \varepsilon_1 > 0$ such that $\mu_0 + \varepsilon_0 < \mu' - \varepsilon_1$ and the following holds

$$\int_{\mu' - \varepsilon_1}^{\mu'} \mu (1 - K(\mu)) dF(\mu) - \int_{\mu_0}^{\mu_0 + \varepsilon_0} \mu \left(K(\mu) - \frac{p}{\mu_0} \right) dF(\mu) = 0 \quad (43)$$

By construction the allocation corresponding to $\hat{K}$ is feasible as it has the same mean as the one corresponding to K , this follows from Equation 43 and Equation 36 and the fact that $q_1(\mu) = \frac{K(\mu)}{1-p} + \text{constant}$ over $[\mu_0, \mu']$ (by Equation 41). By assumption and monotonicity, there must also be values of $\varepsilon_0, \varepsilon_1$ such that $\hat{K} \neq K$. The change in revenue is given by

$$\begin{aligned} \int_{\mu' - \varepsilon_1}^{\mu'} \left[\frac{\mu(m-p)}{(1-p)m} - \frac{1-F(\mu)}{f(\mu)} \right] (1 - K(\mu)) dF(\mu) \\ - \int_{\mu_0}^{\mu_0 + \varepsilon_0} \left[\frac{\mu(m-p)}{(1-p)m} - \frac{1-F(\mu)}{f(\mu)} \right] \left(K(\mu) - \frac{p}{\mu_0} \right) dF(\mu) \end{aligned}$$

which by Equation 43 equals

$$= - \int_{\mu' - \varepsilon_1}^{\mu'} \frac{1-F(\mu)}{f(\mu)} (1 - K(\mu)) dF(\mu) + \int_{\mu_0}^{\mu_0 + \varepsilon_0} \frac{1-F(\mu)}{f(\mu)} \left(K(\mu) - \frac{p}{\mu_0} \right) dF(\mu)$$

Since

$$\frac{1-F(\mu)}{\mu f(\mu)} = \frac{1-\mu}{\mu} \frac{1-F(\mu)}{(1-\mu)f(\mu)}$$

is strictly decreasing on $(0, 1)$, Equation 43 implies that the preceding change in revenue is bounded below by

$$\begin{aligned} \left[\frac{1-F(\mu_0 + \varepsilon_0)}{(\mu_0 + \varepsilon_0)f(\mu_0 + \varepsilon_0)} - \frac{1-F(\mu' - \varepsilon_1)}{(\mu' - \varepsilon_1)f(\mu' - \varepsilon_1)} \right] \\ \times \int_{\mu_0}^{\mu_0 + \varepsilon_0} \mu \left(K(\mu) - \frac{p}{\mu_0} \right) dF(\mu) > 0. \end{aligned}$$

The strict inequality follows from $\mu_0 + \varepsilon_0 < \mu' - \varepsilon_1$ and the positive common weighted mass in Equation 43 for a nontrivial perturbation.

We conclude that a solution to the relaxed problem in the first case has $K(\mu) \in \left\{ \frac{p}{\mu_0}, 1 \right\}$ for all $\mu \in [\mu_0, \mu']$. We can now conclude the proof of the first case in the Proposition. It suffices to note that $K(\mu) = 0$ implies $q_1(\mu) = q_0(\mu) = 0$ and

$K(\mu) = 1$ implies $q_1(\mu) = q_0(\mu) = 1$. Then for the interval $\mu \in [\mu_0, \mu']$ since $K(\mu) = \frac{p}{\mu_0}$ and $q(\mu) = \left(\frac{K(\mu)-p}{1-p}, 1\right)$ (by Equation 41) we obtain $q_1(\mu) = \frac{p}{1-p} \frac{1-\mu_0}{\mu_0}$.

The second case, Equation 42, is treated following similar lines. By definition of μ' , we have $0 \leq K(\mu) \leq 1-p$ for $\mu \in [\mu_0, \mu']$. By arguments analogous to the first case we can show that a solution to the relaxed problem has

$K(\mu) \in \{0, 1-p\}$ for $\mu \in [\mu_0, \mu']$.

To conclude the proof note that for the interval $\mu \in [\mu_0, \mu']$ since $K(\mu) = 1-p$ and $q(\mu) = \left(\frac{K(\mu)}{1-p}, 0\right)$ (by Equation 42) we obtain $q_1(\mu) = 1$. $\square$

D.5 Optimal Mean

Proposition 2 characterizes the structure of the optimal mechanism for the relaxed problem. Using this structure, we can show that the optimal target mean for a given price p is $\mathbf{EH}_0^p$.

Lemma 5. *If $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-increasing then the optimal mean in Equation 14 for a solution to the relaxed problem with price p is $m = \mathbf{EH}_0^p$.*

Proof. Consider for contradiction that q is of one of the forms identified in Proposition 2 but the corresponding target mean $m < \mathbf{EH}_0^p$.

Case I: Let q be of the first form

$$q(\mu) = \begin{cases} (0, 0) & \text{if } \mu \leq \mu_0 \\ \left(\frac{p}{1-p} \frac{1-\mu_0}{\mu_0}, 1\right) & \text{if } \mu \in [\mu_0, \mu'] \\ (1, 1) & \text{otherwise} \end{cases}$$

If $\mu_0 < \mu'$ then we can construct a profitable deviation by choosing $\delta^*, \delta > 0$ and by perturbing the allocation to $(0, 0)$ on $[\mu_0, \mu_0 + \delta^*)$ and to $(1, 1)$ on $[\mu' - \delta, \mu']$. We can choose δ, δ^* such that the new mean of the allocation is $m' \leq \mathbf{EH}_0^p$ and the following holds

$$\int_{\mu_0}^{\mu_0 + \delta^*} \mu \frac{p}{1-p} \frac{1-\mu_0}{\mu_0} dF(\mu) = \int_{\mu' - \delta}^{\mu'} \mu \left(1 - \frac{p}{1-p} \frac{1-\mu_0}{\mu_0}\right) dF(\mu)$$

As $\frac{1-F(\mu)}{\mu f(\mu)}$ is non-increasing, this perturbation reduces the information rent and increases the gross surplus.

If $\mu_0 = \mu'$ then we can create a profitable perturbation by changing the allocation to $(1, 0)$ on $[\mu', \mu' + \delta)$ for small enough $\delta > 0$.

Case II: Let q be of the second form

$$q(\mu) = \begin{cases} (0, 0) & \text{if } \mu \leq \mu_0 \\ (1, 0) & \text{if } \mu \in [\mu_0, \mu'] \\ (1, 1) & \text{otherwise} \end{cases}$$

If $\mu_0 = \mu'$, we can repeat the previous argument.

If $\mu_0 < \mu'$ then we can construct a profitable deviation by choosing $\delta > 0$ and by perturbing the allocation to $(1, 0)$ on $[\mu', \mu' + \delta)$. This is feasible for a small enough δ as $m < \mathbf{E}H_0^p \leq 1$ implies $\mu' < 1$. $\square$

D.6 Proof of Theorem 3

The following lemma isolates the implication of price optimality needed for the first form in Proposition 2.

Lemma 6. *There exists $\varepsilon > 0$ such that, if F has a positive C^1 density f with $\|f - 1\|_\infty < \varepsilon$ and $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-increasing, an optimal relaxed mechanism of the first form in Proposition 2, with $\mu_0 < \mu'$ and conditional mean $\mathbf{E}H_0^p$, satisfies*

$$\frac{3}{10} < p < \frac{1}{2}, \quad 0 < p < \mu_0 < \mu' < 1, \quad (44)$$

$$\alpha := \frac{p(1-\mu_0)}{(1-p)\mu_0} > \frac{1}{3}, \quad \mu_0 < \frac{3p}{1+2p}. \quad (45)$$

The proof is in (Chopra and Ely, 2026c, ??).

Proof of Theorem 3. Choose an optimal relaxed mechanism using Proposition 2 and Lemma 5. It has one of the two forms and platform mean $m = \mathbf{E}H_0^p = p(1 - \ln p)$. Its probability of platform access and mean satisfy

$$\mathbf{E}[\bar{q}] = \int_0^1 \bar{q}(t) dF(t) > 0, \quad (46)$$

$$\int_0^1 tq_1(t) dF(t) = m\mathbf{E}[\bar{q}]. \quad (47)$$

The envelope payments and attainment are detailed in (Chopra and Ely, 2026c, ??). By Corollary 1, it suffices to show that G_q , the belief law induced by platform access alone, is a mean-preserving contraction of H_0^p .

First note that f is non-decreasing. The weighted inverse hazard tends to one as t tends to one, so its monotonicity implies $1 - F(t) \geq (1 - t)f(t)$. Differentiating its

logarithm gives

$$\frac{f'(t)}{f(t)} \geq \frac{1}{1-t} - \frac{f(t)}{1-F(t)} \geq 0, \quad 0 < t < 1. \quad (48)$$

The second form. Consider the second form in [Proposition 2](#). Types below μ_0 are excluded; types in $[\mu_0, \mu')$ receive platform access only when their valuation is one, so $(q_1, q_0) = (1, 0)$; and types in $[\mu', 1]$ receive $(q_1, q_0) = (1, 1)$. Platform full participation in [Equation 6](#) implies $\mu' \geq p$. For $z < 1$,

$$G_q(z) = \begin{cases} 0, & z < \mu', \\ \frac{F(z) - F(\mu')}{\mathbf{E}[\bar{q}]}, & \mu' \leq z < 1. \end{cases} \quad (49)$$

The omitted mass is an atom at one. On $(\mu', 1)$,

$$\frac{d}{dz}[G_q(z) - H_0^p(z)] = \frac{z^2 f(z) - p \mathbf{E}[\bar{q}]}{z^2 \mathbf{E}[\bar{q}]}. \quad (50)$$

The numerator in [Equation 50](#) is strictly increasing, so the CDF difference can cross zero only from below. Since it starts nonpositive and its integral at one is zero by [Equation 47](#), its integral up to every z is nonpositive. This proves the required convex order and hence implementability by [Corollary 1](#). The argument also covers $\mu' = p$ and $\mu_0 = \mu'$.

The first form. In the first form of [Proposition 2](#), types below μ_0 are excluded, types in $[\mu_0, \mu')$ receive $(q_1, q_0) = (\alpha, 1)$, and types in $[\mu', 1]$ receive $(q_1, q_0) = (1, 1)$, with α as in [Equation 45](#). The posterior induced by platform access alone is

$$v_q(t) = \begin{cases} \frac{\alpha t}{1 - (1 - \alpha)t}, & \mu_0 \leq t < \mu', \\ t, & \mu' \leq t \leq 1. \end{cases} \quad (51)$$

The first range of posteriors starts at $v_q(\mu_0) = p$ and ends at $\frac{\alpha \mu'}{1 - (1 - \alpha)\mu'}$; the second starts at μ' . Between these posterior thresholds, G_q is flat while H_0^p increases. Thus a crossing on the range induced by $[\mu_0, \mu')$ can be reversed across this gap, with a further crossing on the range induced by $[\mu', 1]$. We must therefore verify the integrated CDF inequalities directly.

If $\mu_0 = \mu'$, the two forms coincide and the preceding argument applies. Otherwise, [Lemma 6](#) gives $\alpha > 1/3$. Its proof first compares revenue with a feasible second-form mechanism to rule out extreme prices, then combines the price and cutoff

first-order conditions to obtain this bound. We use it below to control the curvature of the integral difference.

Conditional on platform access, types have distribution $\frac{\bar{q}(t) dF(t)}{\mathbf{E}[\bar{q}]}$. Here $\bar{q}(t) = 1 - (1 - \alpha)t$ on $[\mu_0, \mu']$ and $\bar{q}(t) = 1$ on $[\mu', 1]$. Applying Equation 51 gives G_q , with no atom at p . The target H_0^p in Equation 12 has CDF $1 - \frac{p}{z}$ on $[p, 1)$ and an atom of size p at one. Both laws have mean m by Equation 47.

Write $[x]_+ = \max\{x, 0\}$. For $p \leq z \leq 1$, the integrated CDF is the expectation of $[z - \nu_q]_+$ under this conditional distribution. Hence

$$\mathbf{E}[\bar{q}] \int_p^z G_q(v) dv = \int_0^1 [z\bar{q}(t) - tq_1(t)]_+ dF(t). \quad (52)$$

For $t \in [\mu_0, \mu']$ the integrand in Equation 52 equals $[z(1 - t) - (1 - z)\alpha t]_+$. For $t \geq \mu'$, it equals $[z - t]_+$, and

$$[z - t]_+ \leq [z(1 - t) - (1 - z)\alpha t]_+, \quad (53)$$

since the second expression inside the positive part exceeds the first by $(1 - \alpha)t(1 - z) \geq 0$. Using this upper bound for every $t \geq \mu_0$, and observing that its integrand is positive exactly when $t < \frac{z}{\alpha + (1 - \alpha)z}$, yields

$$\mathbf{E}[\bar{q}] \int_p^z G_q(v) dv \leq \int_{\mu_0}^{\frac{z}{\alpha + (1 - \alpha)z}} [z(1 - t) - (1 - z)\alpha t] dF(t). \quad (54)$$

The integrated CDF of H_0^p is

$$\mathbf{E}[\bar{q}] \int_p^z H_0^p(v) dv = \mathbf{E}[\bar{q}] [z - m - p \ln z]. \quad (55)$$

Define its difference from the upper bound in Equation 54 by

$$D(z) := \mathbf{E}[\bar{q}] [z - m - p \ln z] - \int_{\mu_0}^{\frac{z}{\alpha + (1 - \alpha)z}} [z(1 - t) - (1 - z)\alpha t] dF(t). \quad (56)$$

If $D(z) \geq 0$, this upper bound on the integrated CDF of G_q is itself bounded above by Equation 55. Thus the required convex-order inequalities follow once we show

$$D(z) \geq 0, \quad p \leq z \leq 1. \quad (57)$$

Apply Leibniz's rule to Equation 56. Its integrand vanishes at the upper limit, so the moving limit contributes no boundary term to the first derivative:

$$D'(z) = \mathbf{E}[\bar{q}] \left(1 - \frac{p}{z}\right) - \int_{\mu_0}^{\frac{z}{\alpha + (1 - \alpha)z}} [1 - (1 - \alpha)t] dF(t). \quad (58)$$

At $z = p$, Equation 45 gives $\frac{p}{\alpha+(1-\alpha)p} = \mu_0$, so Equation 56 and Equation 58 give $D(p) = D'(p) = 0$.

At $z = 1$, the upper limit is one and the integrand in Equation 56 is $1 - t$. To evaluate this endpoint, subtract Equation 47 from Equation 46 and use $q_0(t) = 1$ for $t \geq \mu_0$ and zero below:

$$\begin{aligned} (1 - m)\mathbf{E}[\bar{q}] &= \int_0^1 (1 - t)q_0(t) dF(t) \\ &= \int_{\mu_0}^1 (1 - t) dF(t). \end{aligned} \quad (59)$$

Both sides are the unconditional probability of platform access with valuation zero. Substitution in Equation 56 gives $D(1) = 0$.

The upper limit in Equation 58 has derivative $\frac{\alpha}{[\alpha+(1-\alpha)z]^2}$, and $1 - (1 - \alpha)t$ evaluated there is $\frac{\alpha}{\alpha+(1-\alpha)z}$. Multiplying these terms by the density at the upper limit gives the second term in

$$D''(z) = \frac{p\mathbf{E}[\bar{q}]}{z^2} - \frac{\alpha^2 f\left(\frac{z}{\alpha+(1-\alpha)z}\right)}{[\alpha + (1 - \alpha)z]^3}. \quad (60)$$

In Equation 60, divide the positive term being subtracted by the first positive term. Substituting $z = \frac{\alpha t}{1-(1-\alpha)t}$ gives the ratio

$$\frac{\alpha t^2 [1 - (1 - \alpha)t] f(t)}{p\mathbf{E}[\bar{q}]}. \quad (61)$$

Although $1 - (1 - \alpha)t$ decreases with t , its product with t^2 is strictly increasing: the bound in Lemma 6 implies

$$\frac{d}{dt} \{t^2 [1 - (1 - \alpha)t]\} = t[2 - 3(1 - \alpha)t] \geq t(3\alpha - 1) > 0. \quad (62)$$

By Equation 48, f is positive and non-decreasing. Hence the ratio in Equation 61 increases strictly with t . The substitution $z = \frac{\alpha t}{1-(1-\alpha)t}$ is strictly increasing on $[\mu_0, 1]$ as well. Thus, once the subtracted term in Equation 60 exceeds the first term, it continues to do so: D'' can change sign only from positive to negative.

It follows that D is convex on $[p, z^*]$ and concave on $[z^*, 1]$ for some $z^* \in [p, 1]$, allowing either portion to be empty. On the convex portion, D lies above its horizontal tangent at p , since $D(p) = D'(p) = 0$, so $D \geq 0$. On the concave portion, D lies above the chord joining the already nonnegative value $D(z^*)$ to $D(1) = 0$.

That chord is nonnegative, so D remains nonnegative. This proves [Equation 57](#), and therefore

$$\int_p^z [G_q(v) - H_0^p(v)] dv \leq 0 \quad (p \leq z \leq 1), \quad (63)$$

with equality at one by [Equation 47](#). The theorem follows from [Corollary 1](#). $\square$